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# Paleomagnetism and Rock Magnetism of Quaternary Volcanic Rocks and Late Paleozoic Strata, VC-1 Core Hole, Valles Caldera, New Mexico, With Emphasis on Remagnetization of Late Paleozoic Strata

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Paleomagnetic and rock magnetic data obtained from azimuthally unoriented core samples, collected at approximately 1- to 3-m intervals, of Continental Scientific Drilling Program core hole VC-1 have prompted reinterpretations of the Quaternary volcanic stratigraphy intersected by the bore and have aided in evaluating the thermal regime within late Paleozoic strata attending fluid circulation and mineral deposition during and after development of the Toledo and Valles calderas. The results from Quaternary units (Banco Bonito Obsidian: $I = +35.4°$, $a_{95} = 2.8°$ (inclination only determinations), $n = 33$; Battleship Rock Tuff: $D = 359.6°$, $I = +42.4°$, $a_{95} = 2.8°$, $n = 5$ site means (surface sites); VC-1 Rhyolite: $I = +39.2°$, $a_{95} = 12.8°$, $n = 7$; Upper VC-1 Tuff: $I = +37.2°$, $a_{95} = 10.7°$, $n = 13$; Middle VC-1 Tuff: $I = +42.1°$, $a_{95} = 2.1°$, $n = 39$; South Mountain Rhyolite: $D = 350.9°$, $I = +49.9°$, $a_{95} = 3.4°$, $n = 10$ (one surface site)) are consistent with isotopic age data, indicating that the entire moat volcanic sequence intersected is less than 650 kyr. Monitoring of natural remanent magnetization (NRM) intensity, NRM directions, directions of magnetizations isolated during progressive demagnetization, median destructive forces, and rock magnetization parameters has identified systematic variations within the thick Banco Bonito Obsidian and VC-1 Tuff units. The Permian Abo Formation, Pennsylvanian to earliest Permian Madera Limestone, and Pennsylvanian Sandia Formation typically contain a moderate positive inclination magnetization component (Abo Formation: $I = +52.2°$, $a_{95} = 7.4°$, $n = 16$; Madera Limestone: $I = +58.4°$, $a_{95} = 2.8°$, $n = 105$; Sandia Formation: $I = +53.9°$, $a_{95} = 4.8°$, $n = 21$); when residing in magnetite, it is usually unblocked in the laboratory by 350°C; when carried by hematite it is unblocked by 550°C. A moderate negative inclination (e.g., Madera and Abo strata: $D = 173.1°$, $I = -46.6°$, $a_{95} = 5.5°$; $n = 47$ samples; assuming a north seeking positive inclination RM providing a useful means of core orientation), oppositely directed magnetization is occasionally isolated and removed at higher laboratory unblocking temperatures (up to 550°C in magnetite-dominated and 600°C in hematite-dominated lithologies). Where present, magnetizations of shallower inclination (e.g., Madera and Abo strata: $D = 160.1°$, $I = -10.9°$, $a_{95} = 4.4°$, $n = 28$) are removed over higher ranges of unblocking temperatures, and these are probably of late Paleozoic age. The moderate negative and positive inclination magnetizations are in all likelihood viscous partial thermoremanent magnetizations (TRMs) (VPTRMs). These were activated at moderate (~300°C) temperatures between 1.45 and 0.97 Ma, attending and following the main stages of Toledo/Valles caldera development during the Matuyama, and at near-present downhole temperatures (~100°–180+°C) during the Brunhes and possibly within the last 10 kyr, respectively. Importantly, the preservation of negative inclination RM components in magnetite-dominated lithologies of the Madera Limestone dictates that temperatures during the Brunhes chron (past 730 kyr) did not exceed approximately 300°–350°C. The inferred temperatures for thermal activation of the VPTRMs, based upon the Middleton and Schmidt thermal activation curves for both magnetite and hematite derived following Walton, agree with independent estimates of thermal activity, using fluid inclusion and K-Ar isotopic age determination data, during the Quaternary. The results further illustrate the potential usefulness of thermal activation theories for inferring temperatures of past geologic events, especially when they can be applied to magnetizations of similar age residing in both magnetite and hematite, and of paleomagnetic and rock magnetic studies in general in continuous coreholes in complex geologic environments.

## INTRODUCTION

It is a rare and most gratifying opportunity to conduct paleomagnetic and rock magnetic research on a sequence of rock units from an area which has been the subject of numerous comprehensive geologic and geophysical studies over several decades and, recently, has received considerable attention as a result of the Continental Scientific Drilling Program (CSDP) [*Heiken*, 1986]. Given the essentially continuous (greater than 98%) recovery of core material in the VC-1 experiment [*Goff et al.*, 1986], this research had three goals. First, paleomagnetic and rock magnetic data may support/refine proposed stratigraphy for the Quaternary volcanic section intersected in the VC-1 core hole. Of note is the fact that preliminary data prompted a reinvestigation of the stratigraphy of the volcanic section, which subsequently showed that the upper and lower Bandelier Tuffs, the main eruptive products of the Toledo and Valles calderas, had not been intersected (F. Goff, personal communication, 1986). Second, the data could also help detect vertical variations in the primary magnetic mineralogy of the volcanic units and subsequent alteration affecting these units. Finally, given the significance of and recent interest in thermoviscous partial to complete remagnetization of sedimentary strata [e.g., *Kent*, 1985; *Dunlop*, 1986; *Jackson and Van der Voo*, 1986; *Kent and Miller*, 1987], the effects of late Tertiary thermochemical processes and, in particular, thermal perturbations on a thick sequence of late Paleozoic strata, whose (primary) magnetization probably resided in both hematite (the Permian Abo Formation) and magnetite (the Pennsylvanian Madera Limestone), were of interest. Paleomagnetic data from these units, in particular laboratory unblocking temperature ($T_{lub}$) spectra of negative in-

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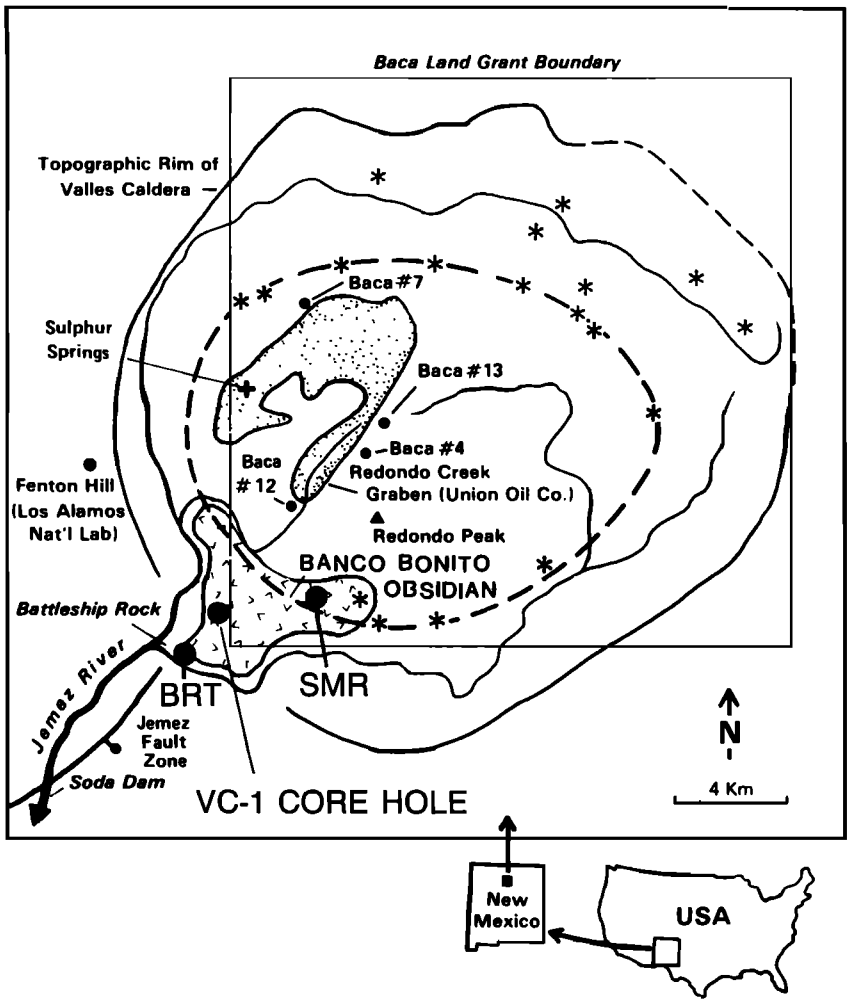



Fig. 1. Location map of Valles caldera and VC-1 core hole; checked pattern shows Banco Bonito obsidian flow; stipple pattern shows area of intense near-surface leaching. Squares denote locations of surface paleomagnetic sampling sites [after *Goff et al.*, 1986].

clination magnetizations in magnetite and hematite, constrain the age of moderate (~300°C) heating of these rocks to before the Brunhes chron (~730 kyr).

This contribution represents the initial phase of paleomagnetic and rock magnetic research on, principally, core hole material from the Valles caldera area (Figure 1). The research has identified numerous additional phases of investigation. Some of these, because of their importance for understanding the geologic history of the Valles caldera, are discussed here, albeit in limited detail. Along with work on recently acquired VC-2A core hole material, continued studies of VC-1 material and nearby surface exposures of Paleozoic strata stratigraphically equivalent to those intersected in VC-1 are in progress in the University of New Mexico laboratory.

## Previous Paleomagnetic Research

The monumental studies by *Doell and Dalrymple* [1966], *Dalrymple et al.* [1967], and *Doell et al.* [1968] on the upper and lower Bandelier tuffs and younger units of the Valles caldera aided in defining the worldwide geomagnetic polarity time scale for Quaternary time [e.g., *Cox*, 1982]. Although orthogonal progressive demagnetization diagrams [*Zijderveld*, 1967] were not used to assess the nature of the natural remanent magnetization (NRM), and specifically the origin of secondary components, in the units investigated, all studies employed rigorous demagnetization techniques to remove secondary, generally present-day field RM components and isolate what were probably thermoremanent magnetizations (TRM). Consequently, their results [*Doell et al.*, 1968, Table 1] for surface units intersected in the VC-1 experiment are useful for comparisons with data from the present contribution.

Initial stratigraphic logging of VC-1 (for example, *Goff* [1985], *Goff et al.* [1984], and *Gardner* [1985] and further summarized by *Goff et al.* [1986]) suggested that the lower volcanic sequence of VC-1 (pre-Battleship Rock Tuff) consists of extremely lithic-rich, welded upper and lower Bandelier Tuff and a lithic-rich welded pre-Bandelier ash flow. Logically, it was anticipated that the rocks intersected, if their primary magnetic mineralogy was preserved, would yield reverse polarity RMs of moderately negative inclination (approximately −34° for the Upper (Tshirege) Member of the Bandelier Tuff and −46° for the Lower (Otowi) Member of the Bandelier Tuff; Table 1). To the contrary, only positive inclinations were isolated in preliminary pilot demagnetization work on core sections from this interval [*Geissman*, 1985], and the concern was raised that these rocks, if not the entire VC-1 section, were remagnetized by the drill string during coring, as reported in previous work on young lavas [e.g., *Van Alstine and Gillett*, 1987] as well as sedimentary rocks [*Van Alstine et al.*, 1985]. This possibility appeared unlikely with continued de-

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magnetization work, numerous discussions with other researchers who have studied bore core from igneous and sedimentary rocks, and review of published results [e.g., *Evans and Maillol*, 1986; *Henry*, 1982; *Kodama*, 1984; *Rosenbaum*, 1986; *Van Alstine and Gillett*, 1987]. Most notably, RM inclinations in individual units appeared statistically distinct, which was subsequently verified with thorough analyses (see below). Inspection of orthogonal demagnetization diagrams often revealed the removal of a very steep positive inclination RM, albeit of relatively small contribution to the NRM, at very low inductions or $T_{lub}$s and thereafter a generally univectorial decay to the origin of a moderate positive inclination RM. The secondary components were probably drilling-induced RMs [e.g., *Burmester*, 1977]. Furthermore, initial paleomagnetic investigations of welded tuffs intersected in VC-2A, thought to be intracaldera Bandelier Tuff, showed that these rocks retained a consistent, moderately negative inclination RM (J. W. Geissman, unpublished data, 1987). Thus the quandary concerning normal polarity TRMs in the Bandelier Tuff remained until comprehensive petrographic examination, followed by detailed major and trace element geochemical work (F. Goff, personal communication, 1986), revealed that the lower tuff section of VC-1 was not the Bandelier Tuff but, rather, a previously unidentified sequence of postcaldera ash flow tuffs now named the VC-1 Tuffs.

Of the Paleozoic formations intersected in VC-1, paleomagnetic data from surface equivalent units are only available for the Permian Abo Formation. To define the limits of the late Paleozoic Kiaman superchron of reverse polarity, *Peterson and Nairn* [1971] sampled the Abo Formation near Abo Pass, 140 km south of the Valles caldera. These workers collected from four to six samples at each of eight sites at the locality, and thus their work was limited in scope. In studies of the rotation of the Colorado Plateau, *Steiner* [1988] has carried out more extensive sampling of the Abo Formation near Abo Pass. Based upon thermal demagnetization experiments, which generally isolated a well-defined magnetization in a $T_{lub}$ interval of 400° to over 645°C, a mean direction of $D = 151.9°$, $I = 6.3°$, $a_{95} = 1.8$, $k = 68$ (6 sites, 68 samples of the 143 collected used to calculated a formation mean) was obtained.

While published paleomagnetic data on surface exposures of the middle Pennsylvanian to earliest Permain (?) Madera Limestone and the Pennsylvanian Sandia Formation do not exist, studies are underway in the University of New Mexico (UNM) laboratory. Based upon Carboniferous paleomagnetic data from North America and published apparent polar wander paths [e.g., *Roy and Morris*, 1983; *Van der Voo*, 1988], a primary magnetization in these strata, if present, would have an inclination less than approximately 12°, assuming an age of RM acquisition between 285 and 320 Ma and pole positions ranging from 40°N, 130°E to 27°N, 122°E.

For all late Paleozoic strata the inclinations of early acquired RMs would necessarily be very shallow and polarity information is of little concern nor readily inferred in dealing with azimuthally unoriented core. Nonetheless, it is expected that primary RMs in the Abo Formation and most of the Madera Limestone, at least to the base of the Westphalian [e.g., *Roy and Morris*, 1983], would be of reverse polarity, acquired during the Kiaman superchron [e.g., *Cox*, 1982].

## FIELD AND LABORATORY TECHNIQUES

Azimuthally unoriented sections of core from VC-1 were obtained from Los Alamos National Laboratory in late 1984 and from the Department of Energy core repository in Grand Junction, Colorado, in 1986 and 1987. Initially, 262 core sections were collected at approximately 3-m intervals for the study. In September 1987, 19 additional core sections were obtained from portions of the late Paleozic section, increasing the sampling density. A breakdown of the core sections with respect to the revised VC-1 stratigraphy is given in Figure 2. The length of each core section ranged from 6 to 18 cm. Each core section is referred to as a separate, independent sample in this contribution. The attitude of the VC-1 borehole is assumed to be within a few degrees of vertical throughout the experiment. The hole was spudded vertically; the goal of extremely high core recovery required that drilling was carried out slowly to maintain a constant borehole orientation. The one orientation survey conducted during the drilling experiment, at 475 m depth, indicated that the core was 1/4° off vertical.

Standard (2.54 cm × 2.3 cm) specimens for paleomagnetic, some rock magnetic, and anisotropy of magnetic susceptibility analyses were prepared by drilling perpendicular to the axis of each core sample, generally (always for the sedimentary rocks) using a entirely nonmagnetic drill bit, at the UNM laboratory. Specimens were cut using nonmagnetic copper/diamond saw blades. To evaluate the possibility of contaminating the NRM with artificial, drilling-related magnetizations in the laboratory, the multiple-coring procedure of *Van Alstine and Gillett* [1982] was carried out on selected core samples from all important lithologies. Some results of this endeavor are shown in Figure 3 and indicate that laboratory drilling appears to have had little effect on the NRM of all lithologies, except, possibly, in the case where NRM intensities are extremely low (e.g., less than $2 \times 10^{-4}$ A/m), and demagnetization results indicate that the NRM is dominated by very low coercivity phases.

In cases where core sections were too fragile to drill, approximately 1.2-cm-thick "wafers" were sliced perpendicular to the core segment and were further sliced into cubes and glued with nonmagnetic cement into plastic boxes for measurement. Where thermal demagnetization was required, cubes were cast into high-temperature alumina cement to fit into plastic boxes each time they were measured. Again, all specimen preparation involved using a nonmagnetic copper/-diamond saw blade.

Abundant conventional specimens of a particular core segment were often difficult to obtain. In some instances, the acquisition of critical rock magnetic or additional paleomagnetic information from a particular interval will require additional sampling of core.

Sampling of surface outcrops in Battleship Rock Tuff (five sites) at Battleship Rock and South Mountain Rhyolite (one site) near Jemez Falls (Figure 1) was carried out to compare surface data with possibly correlative units retrieved in VC-1. A portable field drill, again with nonmagnetic drill bits, was used to collect 7–10 independently oriented samples per site. As our sites were on National Forest land, all drill holes were subsequently patched with a permanent, color-matched cement. Samples were oriented using magnetic as well as, where possible, solar compasses. The volcanic units sampled, based on previous field mapping [*Smith et al.*, 1970] and our own observations, are flat-lying.

Progressive alternating field (af), thermal, and low-temperature [e.g., *Merrill*, 1970] demagnetization techniques were employed to evaluate the NRM of all lithologies. Over 95% of the measurements were made on a computer-interfaced Schonstedt SSM-1a spinner magnetometer, the shaft of which was modified with an optical shaft encoder. Using a 32-spin measurement procedure and evaluating the

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TABLE 1. Paleomagnetic Data From Quaternary Volcanic Rocks and Late Paleozoic Strata, VC-1 Core Hole and Valles Caldera

| Unit | Number of Sites | Number of Samples | In Situ ChRM[a] | | | | | | | VGP Position | | Age, Ma | Comments | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | D, deg | I, deg | Arithmetic Mean[b] | s.d.[b] | R | $a_{95}$ | k | Latitude, °N | Longitude, °E | | | |
| *Quaternary Volcanic Rocks* | | | | | | | | | | | | | | |
| Banco Bonito Obsidian | 1 | 8 | 359.0 | 41.3 | | | 7.726 | 11.2 | 26 | 77.8 | 77.6 | 0.13 ± 0.10[c] | baked contact test, any secondary RM removed by 40 mT | *Doell et al.* [1968] |
| | 1 | 33[d] | ··· | 35.4 | 35.4 | 6.92 | ··· | 2.8 | 69 | ··· | ··· | | | This study |
| Battleship Rock Tuff | 1 | 9 | 358.3 | 43.2 | | | 8.975 | 2.9 | 317 | 79.2 | 81.6 | | inclusion test, any secondary RM removed by 40 mT | *Doell et al.* [1968] |
| Site BTR1 | | 7 | 358.4 | 38.7 | | | 6.979 | 3.1 | 280 | | | | surface site | this study |
| Site BTR2 | | 8 | 357.5 | 44.1 | | | 7.979 | 2.7 | 337 | | | | surface site | this study |
| Site BTR3 | | 8 | 357.9 | 42.8 | | | 7.995 | 1.4 | 1325 | | | | surface site | this study |
| Site BTR4 | | 6 | 3.7 | 44.0 | | | 5.979 | 3.7 | 238 | | | | surface site | this study |
| Site BTR5 | | 8 | 0.2 | 43.3 | | | 7.983 | 2.4 | 412 | | | | surface site | this study |
| Total | | | 359.6 | 42.4 | | | | 2.8 | 750 | 78.2 | 78.0 | | surface sites | this study |
| VC-1 Rhyolite | 1 | 7[d] | ··· | 39.2 | 39.3 | 11.4 | | 12.8 | 26 | ··· | ··· | 0.356 ± 0.061[e] | | this study |
| Upper VC-1 Tuff | 1 | 13[d] | ··· | 37.2 | 37.1 | 14.7 | | 10.7 | 15 | ··· | ··· | | | this study |
| Middle VC-1 Tuff | 1 | 39[d] | ··· | 42.1 | 42.1 | 5.7 | | 2.1 | 101 | ··· | ··· | | | this study |
| Lower VC-1 Tuff | ··· | ··· | ··· | ··· | | | | ··· | ··· | ··· | ··· | | | |
| Total | 1 | 52[d] | ··· | 40.9 | 40.9 | 8.9 | | 2.9 | 41 | ··· | ··· | | | this study |
| South Mountain Rhyolite | 1 | 10 | 357.0 | 56.5 | | | 9.621 | 10.1 | 24 | 87.3 | 191.4 | | | *Doell et al.* [1968] |
| Site 5MR1 | | 10 | 350.9 | 49.9 | | | 9.947 | 3.4 | 169 | 80.9 | 131.3 | | surface site | this study |
| Bandelier Tuff Tshirege Member | 6 | 36 | 173.6 | −33.8 | | | 35.748 | 2.0 | 139 | −71.8 | 273.1 | 1.02 ± 0.04<br>1.06 ± 0.03<br>1.07 ± 0.03<br>1.1 ± 0.1<br>1.19 ± 0.04<br>1.24 ± 0.05 | | *Doell et al.* [1968] |
| Otowi Member | 6 | 34 | 174.0 | −45.9 | | | 33.865 | 1.6 | 244 | −80.0 | 285.4 | 1.37 ± 0.04<br>1.44 ± 0.04<br>1.48 ± 0.09 | | *Doell et al.* [1968] |
| *Late Paleozoic Strata* | | | | | | | | | | | | | | |
| Abo Formation | 6 | 68 | 151.9 | −6.3 | | | | 1.8 | 68 | 49.3 | 119.6 | | | *Steiner* [1988] |
| Brunhes age overprint | 1 | 16[d] | ··· | 52.2 | 52.2 | 11.5 | | 7.4 | 26 | ··· | ··· | | | |
| Matuyama age overprint | 1 | 22[d] | ··· | −35.8 | −35.5 | 9.6 | | 5.0 | 36 | ··· | ··· | | | |
| Late Paleozoic RM | 1 | 19[d] | ··· | −7.4 | −7.4 | 10.7 | | 6.1 | 31 | ··· | ··· | | | |
| Madera Limestone | | | | | | | | | | | | | | |
| Brunhes age overprint | 1 | 102[d] | ··· | 58.4 | 58.4 | 9.97 | | 2.8 | 33 | ··· | ··· | | | this study |
| Matuyama age overprint | 1 | 47[d,f] | 173.1 | −46.6[g] | | | | 5.5 | 18 | −79.5 | 289.5 | 1.1[h] | | this study |
| | 1 | 51[d] | ··· | −47.1 | −47.1 | 8.04 | | 2.9 | 51 | ··· | ··· | | | this study |

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| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Late Paleozoic RM | 1 | 28[d,f] | 160.1 | −10.9[g] | | | 4.4 | 41 | 54.9 | 114.1 | this study |
| | 1 | 28[d] | ... | −9.5 | −9.5 | 8.26 | 3.9 | 48 | ... | ... | this study |
| Sandia Formation | | | | | | | | | | | |
| Brunhes age overprint | 1 | 21[d] | ... | 53.9 | 53.9 | 8.35 | 4.8 | 47 | | | this study |
| Late Paleozoic RM | 1 | 2[d] | ... | 11.6 | | | ... | ... | ... | ... | this study |

[a]Characteristic RM determined from progressive demagnetization; only inclination information is given for azimuthally unoriented VC-1 core segments.
[b]Arithmetic mean and standard deviation (s.d.) for inclination only data.
[c]*Marvin and Dobson* [1979].
[d]VC-1 core material.
[e]*Goff et al.* [1986].
[f]Thirty one from Madera Limestone, 16 from Abo Formation.
[g]Mean direction determined assuming northward declination of the low $T_{lub}$, low-coercivity magnetization of moderate positive inclination.
[h]*Ghazi and Wampler* [1987].
[i]Nineteen from Madera Limestone, 7 from Abo Formation.

signal based on 250 data points counted per rotation, the noise level of the magnetometer is less than approximately $10^{-9}$ A/m$^2$ ($1.0 \times 10^{-6}$ emu). Angular standard deviations calculated from multiple measurements of specimens having total moments ranging from 1.0 to $5.0 \times 10^{-9}$ A/m$^2$ are typically less than 4°. A 2G Enterprises model 760R horizontal access, three-axis superconducting rock magnetometer, with peak to peak noise levels less than $3 \times 10^{-11}$ A/m$^2$ on all axes, was employed for more recent measurements.

Detailed (e.g., 15 or more steps) "pilot" demagnetization procedures were carried out on representative specimens of all lithologies. For af demagnetization, a Schonstedt GSD-1 single-axis demagnetizer modified to allow for the capability of tuning the solenoid and providing peak magnetic inductions (B) up to approximately 100 mT (field, 1000 Oe) was used. In thermal demagnetization, using a Schonstedt TSD-1 unit, treatments were carried out until at least 99% of the NRM had been unblocked. In some cases, the noise level of the magnetometer was exceeded before this level of demagnetization. Nonetheless, RM components were still able to be defined. Peak temperature reached in thermal demagnetization varied with lithology, but for all volcanic rocks it exceeded 620°C (to evaluate the possibility of significant components of the RM residing in hematite or ilmenohematite). For the hematitic strata of the Abo Formation, peak temperature exceeded 675°C, while temperatures less than 500°C were often sufficient to reduce the NRM of Madera Formation carbonate rocks to below the noise level of the spinner magnetometer (at which point demagnetization was halted) and/or unblock over 95% of the NRM. Because the NRM was often unblocked over a narrow (less than 25°C) range of laboratory unblocking temperatures ($T_{lub}$), specimens were kept in a constant position but random orientation in the furnace for each heating run. Specimens were held at temperature for effectively 50 min after the desired temperature had been reached and were subsequently cooled in a magnetic induction of less than 5 nT. Over 90% of the measurement and demagnetization experiments for this contribution were carried out with all instrumentation inside a magnetically shielded room. Magnetic induction in the room, with the exception of proximity to ferrimagnetic portions of instruments, is less than 200 nT.

Several rock magnetic techniques were used to evaluate the carriers of the NRM in each lithology. Plots of high-induction (approximately 0.15 T) magnetization versus temperature to infer Curie temperatures were obtained using a horizontal Curie balance. During heating and cooling, a slow flow of argon was passed over the specimen in the furnace assembly to minimize changes in the magnetic mineralogy. Isothermal remanent magnetization (IRM) acquisition and backfield direct-induction demagnetization curves were obtained using an impulse magnet capable of producing peak inductions of 1.36T. Anhysteretic RM (ARM) was imparted in a dc induction of approximately 0.07 mT in a peak alternating induction of 98 mT.

Components of magnetization were identified on standard orthogonal demagnetization plots [*Zijderveld*, 1967] assuming core segment axes to be vertical. As this contribution provides the first account of paleomagnetic data from VC-1 units, many examples of demagnetization behavior, particularly for late Paleozoic strata, are shown. Directions of RM components were determined from the demagnetization data using the least squares method based on principal component analysis (PCA), modified from *Kirschvink* [1980]. Only components

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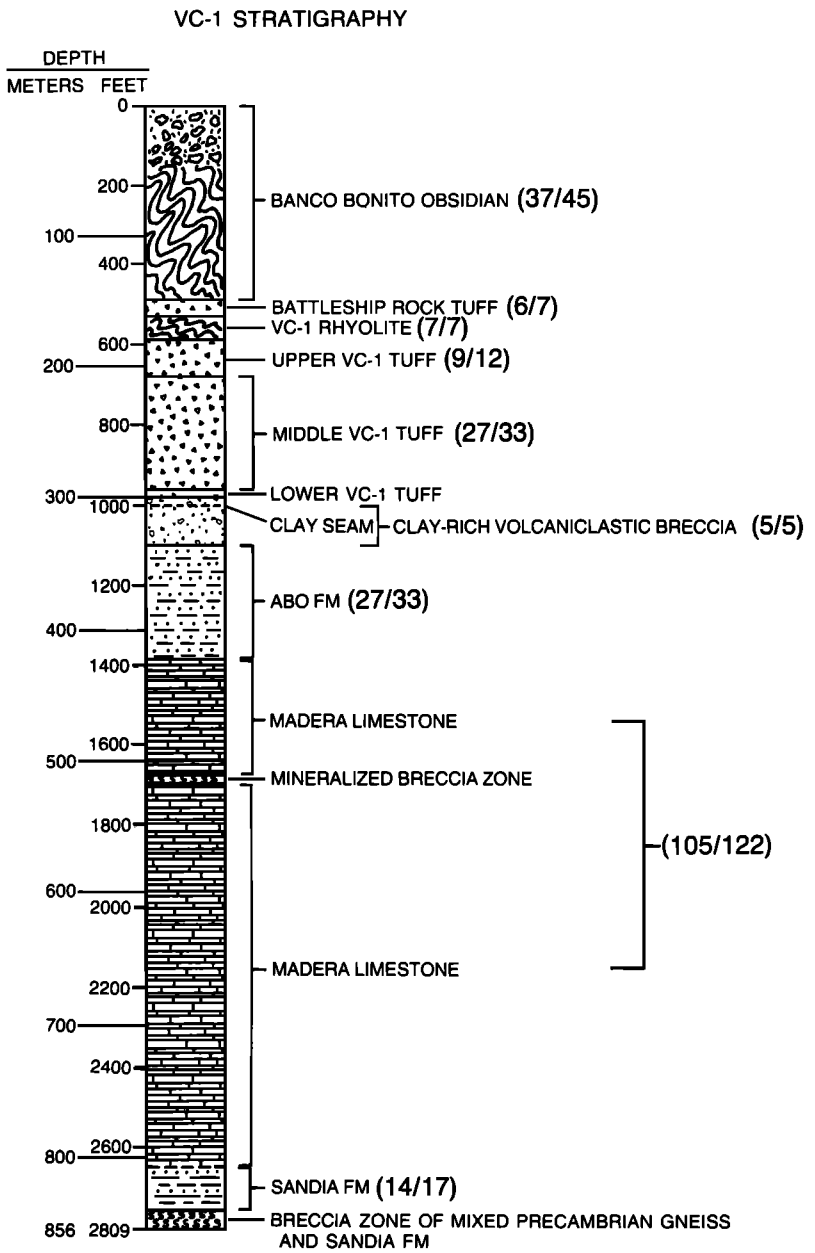



Fig. 2. Revised stratigraphy of VC-1, Valles caldera, New Mexico [after *Goff et al.*, this issue]. Ratio in parentheses for each unit investigated refers to the number of core segments (samples) prepared into specimens for measurement to the total collected.

with mass analyzer detector (MAD) values, determined from minimally three but usually four or more data points, less than 15° (for over 80% of the cases less than 5°) are reported. For all VC-1 material, statistical evaluation of inclination results, (i.e., directions determined with PCA and not those measured after some conveniently predetermined level of demagnetization) was based on the approach of *McFadden and Reid* [1982]. For surface sites and where azimuthal core orientations were inferred using directions of geologically young magnetizations, the method of *Fisher* [1953] was used to compute mean directions and associated statistics.

Bulk magnetic susceptibility (MS) and anisotropy of magnetic susceptibility (AMS) of all volcanic units were measured on a Sapphire SI-2 induction coil susceptibility meter. In general, counting times were 4 s and six orientations were repeated for determination of the standard deviation of the measurement.

Following *Shive* [1986], all measurements of magnetization are expressed in SI units of induction (B, tesla) and magnetic dipole moment/unit volume (M, amperes per meter). An induction of 1 T equates to a cgs magnetic field strength of $10^4$ Oe; a magnetic dipole moment per unit volume of 1 A/m equates to a cgs magnetization of $10^{-3}$ emu/cm$^3$.

## PALEOMAGNETIC AND ROCK MAGNETIC DATA

The paleomagnetic and rock magnetic data are described in reverse stratigraphic order, from youngest to oldest units intersected. The data have been divided into Quaternary volcanic rocks and late Paleozoic sedimentary strata (Figures 4 and 5, respectively).

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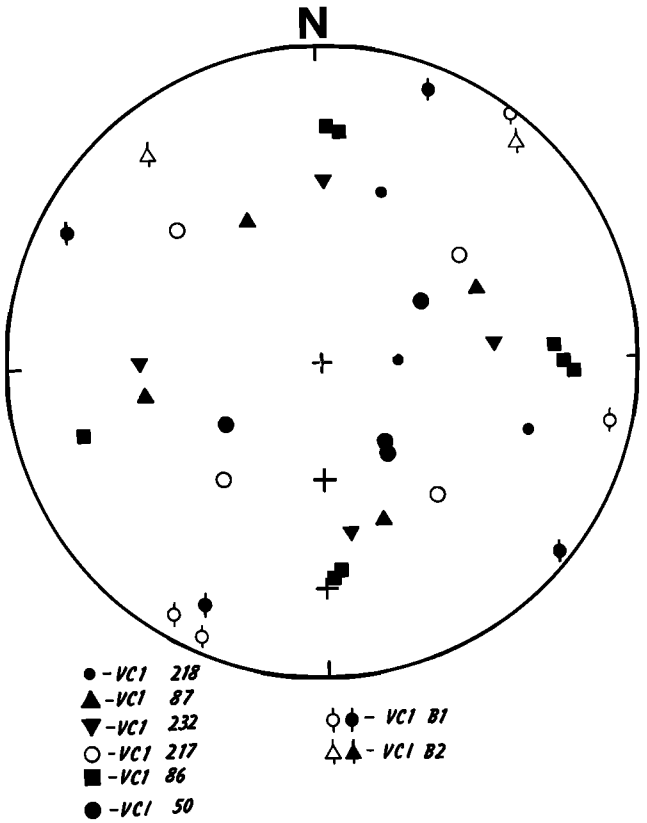



Fig. 3. Equal-area projection of NRM directions of multiple specimens from samples (individual core segments) which were cored in the laboratory in up to four orientations, each separated by approximately 90°. Each symbol represents specimens from one sample; each data point represents one NRM direction. With the exception of specimens from sample VC-218 (Madera Limestone with NRM intensities less than $2 \times 10^{-4}$ A/m), NRM directions are orthogonal (and follow the pattern of coring in the laboratory) and also of constant inclination.

### *Quaternary Volcanic Rocks*

*Banco Bonito Obsidian.* The Banco Bonito Obsidian ($0.13 \pm 0.1$ Ma), consisting of two flow units, was intersected over an interval of 0–149 m. Textural zonation within this unit has been described by *Manley and Fink* [1987]. Specimens from a total of 37 core segments were prepared (Figure 4). NRM intensities range from 0.6 to 70 A/m, and NRM inclinations, all of positive sign, are typically between 20° and 50°. Bulk low-induction magnetic susceptibilities range from $5.0 \times 10^{-3}$ to $2.0 \times 10^{-2}$. Percent lineations of MS do not exceed 4.0, while percent foliations of MS are more variable and range from 1.0 to greater than 10.

In demagnetization, examples of which are shown in Figure 6, specimens from all samples exhibited straightforward behavior. Typically, only a single, moderate positive inclination RM is isolated. Steep positive inclination RMs of low coercivity are occasionally superimposed on what is presumed to be a TRM (e.g., specimens VC1-235a, VC1-65a, VC1-220a; Figure 6). Median destructive inductions generally exceed 50 mT; consequently, thermal demagnetization was required to demonstrate convincingly that this component did decay linearly to the origin and comprise the vast majority of the NRM. Changes in inclination with progressive unblocking (e.g., shallowing of inclination due to subblocking compaction [*Rosenbaum*, 1986]) over wide ranges of unblocking temperatures were not observed, and the implications are discussed in further detail below. The high median destructive inductions, results of detailed thermal demagnetization showing a distributed range of $T_{lub}$, and IRM acquisition data (Figure 7) all suggest that the TRM in the Banco Bonito Obsidian is carried largely by fine, single-domain (SD) and psuedo-single domain (PSD) magnetite grains. The unblocking temperature spectra distributed down to temperatures of 250°C probably reflect an abundance of extremely fine SD particles [e.g., *Dunlop*, 1973, 1981].

The mean inclination of the Banco Bonito Obsidian, determined using inclination values extracted from principal component analysis (Figure 4) is +35.4° ($a_{95} = 2.8°$, $k = 69$). This value is statistically identical, at the 95% confidence level, to the inclination of 41.3° reported for the one site mean by *Doell et al.* [1986] (Table 1). Evaluation of the inclination data on the basis of two flow units (assuming the flow contact at approximately 94.5 m) indicates that both have statistically indistinguishable inclination values.

With respect to vertical variations in magnetic properties in the Banco Bonito Obsidian, it is notable that NRM intensities for core segments in the upper Banco Bonito flow increase to a maximum at a depth of about 61 m, within 34 m of the base of the flow, and then decrease. Inclinations of the TRM display the opposite behavior, while median destructive inductions decrease through the flow unit. The lower obsidian unit exhibits less pronounced behavior, with increases in median destructive induction and (slight) increase in inclination. Bulk MS is essentially constant through the unit, while both percent lineation and foliation of MS decrease with depth.

*Battleship Rock Tuff.* The inferred Battleship Rock Tuff intersected in VC-1 is only 12 m thick (from 149 to 161 m) and is very poorly welded and lithic rich. This horizon of porous pumice in corehole also served as an aquifer for shallow groundwaters, and low-temperature alteration is evident. Consequently, standard specimens could not be prepared. NRM intensities of 3 $cm^3$ cube specimens ranged from 0.1 to 4.0 A/m. RM directions obtained in demagnetization are often anomalously shallow and of mixed inclination.

For comparisons with data obtained from VC-1 material, samples were collected at five sites, vertically spaced over 40 m, in the Battleship Rock Tuff at its type locality where it is sintered to very densely welded. The data obtained are considerably different from those from VC-1 core segments. NRM intensities are relatively high and have a narrow range in value (6.3–2.1 A/m). The direction of both NRM and what we infer to be TRMs are of moderate positive inclination (e.g., specimens BRT1Aa, BRT1Gb, and BRT5Ba, Figure 8 and Table 1). With declination data available, our results are statistically indistinguishable, at a 99% confidence level [*McFadden and Lowes*, 1981], from the single-site mean direction reported by *Doell et al.* [1968]. No systematic variations in magnetic properties from the surface exposures of the Battleship Rock Tuff were observed.

Clearly, the magnetization properties of the Battleship Rock Tuff material intersected in VC-1 differ from those of samples from Battleship Rock Tuff. The possibility exists that the VC-1 material simply does not possess a TRM.

*VC-1 Rhyolite.* This previously unobserved weakly porphyritic obsidian unit ($0.356 \pm 0.061$ Ma, as reported by *Goff et al.* [1986]) was cored over approximately 19 m (from 161 to

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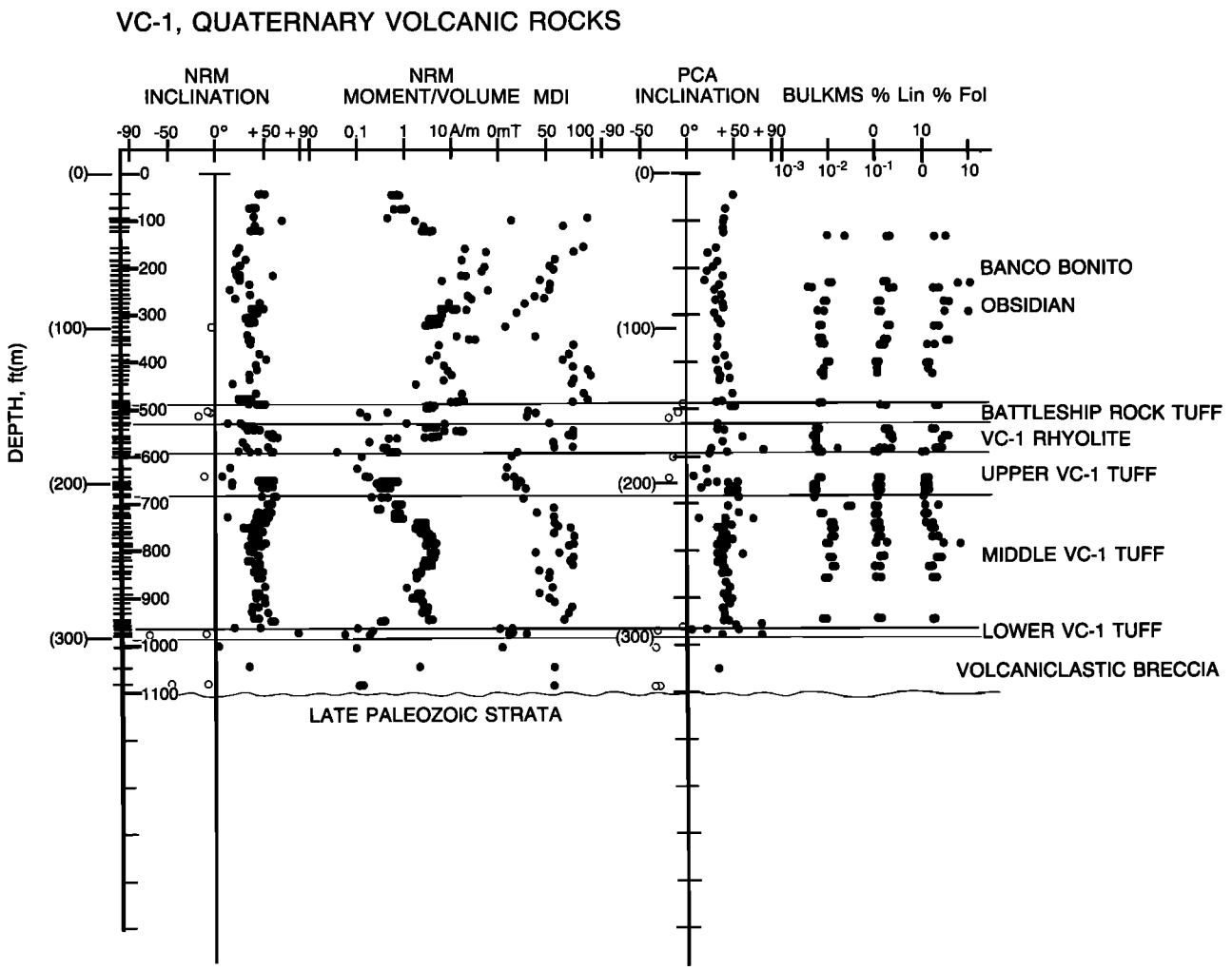



Fig. 4. Paleomagnetic and rock magnetic data from the upper 335 m of VC-1, including all Quaternary volcanic units. Data include NRM inclinations, NRM intensities, median destructive inductions (MDI), and inclinations of RM components determined using principal component analysis, bulk low field magnetic susceptibility (MS, SI units), percent lineation of MS (defined as $100 \times \{k_1 - k_2\}/\{[k_1 + k_2 + k_3]/3\}$, where $k_1$, $k_2$, and $k_3$ are the maximum, intermediate, and minimum principal susceptibility axes), and percent foliation of MS (defined as $100 \times \{[k_1 + k_2] \times 0.5 - k_3]/\{[k_1 + k_2 + k_3]/3\}$). Here and in Figure 5, individual tic marks refer to vertical position of core segments studied. Open circles are used to highlight negative inclination directions or RM components.

180 m). Specimens from a total of eight core segments were prepared. NRM intensities varied from 0.2 to 20 A/m, similar to the range observed in the Banco Bonito Obsidian. Bulk MS values ranged between $4 \times 10^{-3}$ and $7 \times 10^{-3}$. The unit exhibits strong MS foliation and lineation.

Demagnetization consistently removed a moderate positive inclination RM of a high median destructive induction and widely dispersed $T_{lub}$. As with the Banco Bonito Obsidian, demagnetization results (e.g., specimen VC1-35a, Figure 8) and IRM acquisition tests suggest that the well-defined RM, in all likelihood a TRM, resides in a spectrum of fine magnetite particles. The eight samples from the VC-1 Rhyolite yield a mean inclination of $\pm 39.2°$, which is albeit poorly determined ($a_{95} = 12.8°$, $k = 26$). The moderate positive inclination is in keeping with the post-600 kyr age of the Quaternary volcanic sequence intersected.

*VC-1 Tuff.* The VC-1 Tuff consists of three cooling units and was intersected between 180 and 299 m. A total of 43 core segments were investigated for the study. NRM intensities range from 0.06 to 4.0 A/m; throughout the interior, well-welded portion of the unit, they range from 1.0 to 4.0 A/m. NRM inclinations generally are between 35° and 65°. Bulk MS values range from $4 \times 10^{-3}$ to $2 \times 10^{-2}$. MS percent foliation values are usually less than 2, while percent lineation values range up to 5.

Again, demagnetization is very straightforward, with occasional low-coercivity RM components of steep positive inclination (e.g., specimens VC1-27a and VC1-103b, Figure 9) removed before the linear decay of the majority of the NRM. Median destructive inductions typically exceed 40 mT throughout the well-welded portion of the unit, necessitating detailed thermal demagnetization treatments to convincingly demonstrate that an RM component, again inferred to be a TRM, decayed in a linear fashion to the origin (e.g., specimens VC1-6a/6b, VC1-7a/7d, and VC1-51a/51b, Figure 9). Less than 5% of this component exhibited $T_{lub}$ above 580°C. While a low Ti-magnetite carries the majority of the component, the fact that an RM component of identical direction is unblocked above 580°C indicates that hematite, which probably formed under deuteric conditions, is also a carrier of the RM.

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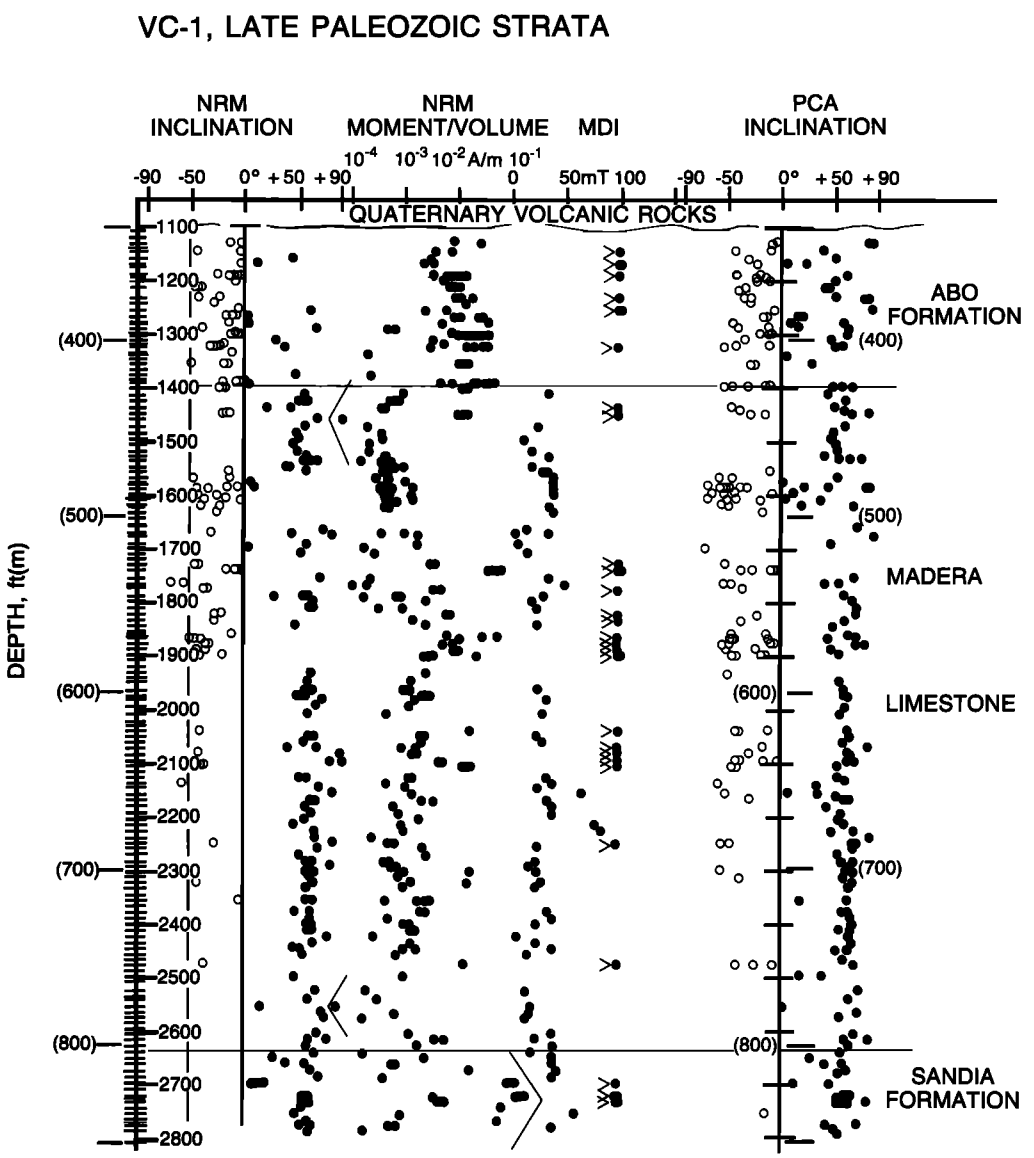



Fig. 5. Paleomagnetic and limited rock magnetic data from the lower 520 m of VC-1, including late Paleozoic strata of the Abo Formation, Madera Limestone, and Sandia Formation. Conventions follow those in Figure 4. No susceptibility data are available from the late Paleozoic strata.

The VC-1 Tuff displays significant internal variations in magnetization properties. NRM intensities progressively increase from the uppermost portion of the unit to a depth of 244 m, then decrease to about 288 m and subsequently increase to the base of the Middle VC-1 Tuff, at approximately 289 m depth. In the Lower VC-1 Tuff NRM intensities are generally an order of magnitude lower than those in the Middle VC-1 Tuff. This unit is so thin that it shows very little welding. Because it overlies clay-rich volcaniclastic breccia, the porous pumice has served as an aquifer, and low-temperature alteration is again evident. Median destructive inductions essentially follow the NRM behavior with maximum values of approximately 80 mT over the interval between 235 and 253 m. Bulk MS values increase from $6 \times 10^{-3}$ to over $2 \times 10^{-2}$ over the upper interval where NRM intensities increase. MS lineation values vary slightly with depth, while foliation values in the middle portion of the tuff are considerably higher than those in the upper part.

Over the depth interval where NRM intensities and median destructive inductions vary considerably, inclinations of RM components remain essentially constant, although a slight inclination decrease occurs where median destructive inductions are at maximum values. In the poorly welded Lower VC-1 Tuff, however, inclinations are both positive and negative in sign. A TRM may not be preserved in this portion of the tuff.

Mean inclination values and associated statistics are given in Table 1. We note that the overall mean inclination of the VC-1 Tuff (+40.9°) is statistically indistinguishable from that of the five surface sites in the Battleship Rock Tuff.

*Volcaniclastic breccia.* Underlying the VC-1 Tuff between 299 and 305 m is a sequence of volcaniclastic breccia deposits. These lithologies were difficult to prepare into specimens, and a limited amount of varied data is presently available. Inclinations of magnetization components from some core segments are negative in sign, but the fidelity of these magnetizations as field recorders is, by nature of the lithologies, in question.

Petrographic, geochemical, and K-Ar isotopic age data (J.

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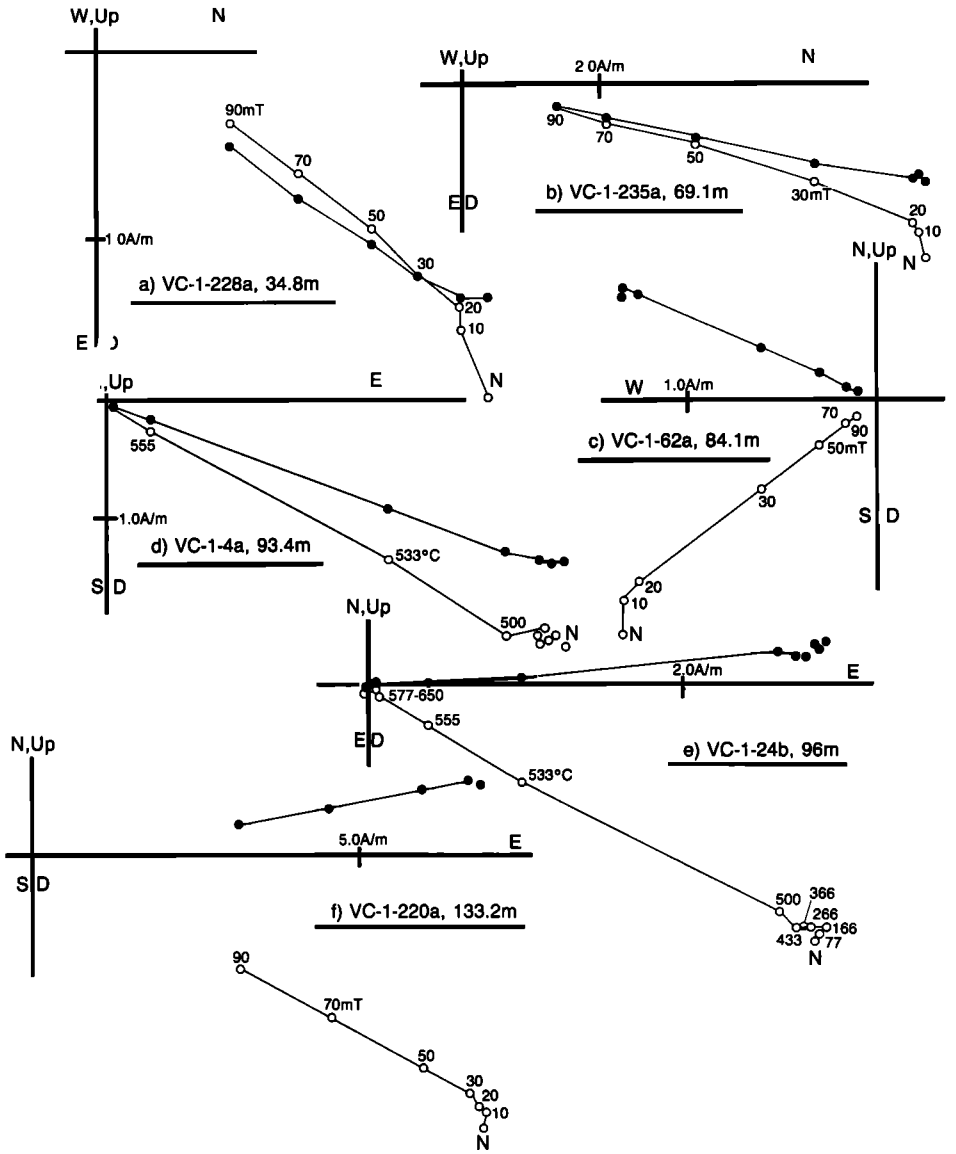



Fig. 6. Representative orthogonal demagnetization diagrams [*Zijderveld*, 1967] for specimens from core segments of the Banco Bonito Obsidian, arranged in stratigraphic order. Such diagrams, here and in Figures 8, 9, 10, 13, and 14 are used to determine the number and directions (using principal component analysis) of magnetization components comprising the NRM and the ranges of peak demagnetization induction and/or laboratory unblocking temperature over which the components are removed. In each diagram the endpoint of the magnetization is simultaneously projected onto the (artificial, with unoriented core) horizontal plane (solid symbols) and a (real) vertical plane (open symbols) assuming the core segment axis is vertical. Peak demagnetization inductions or temperatures are given along the vertical projections. Magnetization intensity (in amperes per meters) is indicated by the a tic along one of the axes.

N. Gardner and F. Goff, unpublished manuscript, 1988) indicate a porphyritic rhyolite in the bottom of this unit is South Mountain Rhyolite (0.494 ± 0.015 Ma). Paleomagnetic data from surface exposures of this unit (Table 1) are consistent with an age of emplacement during the Brunhes chron.

*Late Paleozoic Sedimentary Rocks*

In the interpretation of paleomagnetic data from the late Paleozoic section intersected in VC-1 it must be recognized that sedimentary bedding over portions of the section exhibits moderate dips. As determined, for example, by inspection of oriented core at approximately 475 m, Madera Formation strata dip 25° to the southeast [*Goff et al.*, 1986]. VC-1 was cored into the buried northwest (upthrown) side of the Jemez fault zone (Figure 1), and *Goff et al.* [1986] suggested that tilting of parts of the Paleozoic section encountered as well as the anomalous thickness of the Madera Formation may be due to fault motion along this zone. *Aldrich and Laughlin* [1984] postulated that the Jemez fault zone may be a reactivated segment of a Precambrian or younger structural boundary more recently expressed as the Jemez Lineament. Inspection of core segments by the author during specimen preparation revealed that the amount of tilting is variable and, in particular, strata of the Abo Formation are nearly horizontal although locally fractured and veined, assuming, as previously discussed, an essentially vertical orientation of the borehole. Because the volcanic section intersected in VC-1 is also horizontal and virtually unfractured, tilting and brecciation of any Paleozoic strata adjacent to the fault zone occurred prior to 0.5 Ma, during late Tertiary and Quaternary deformation.

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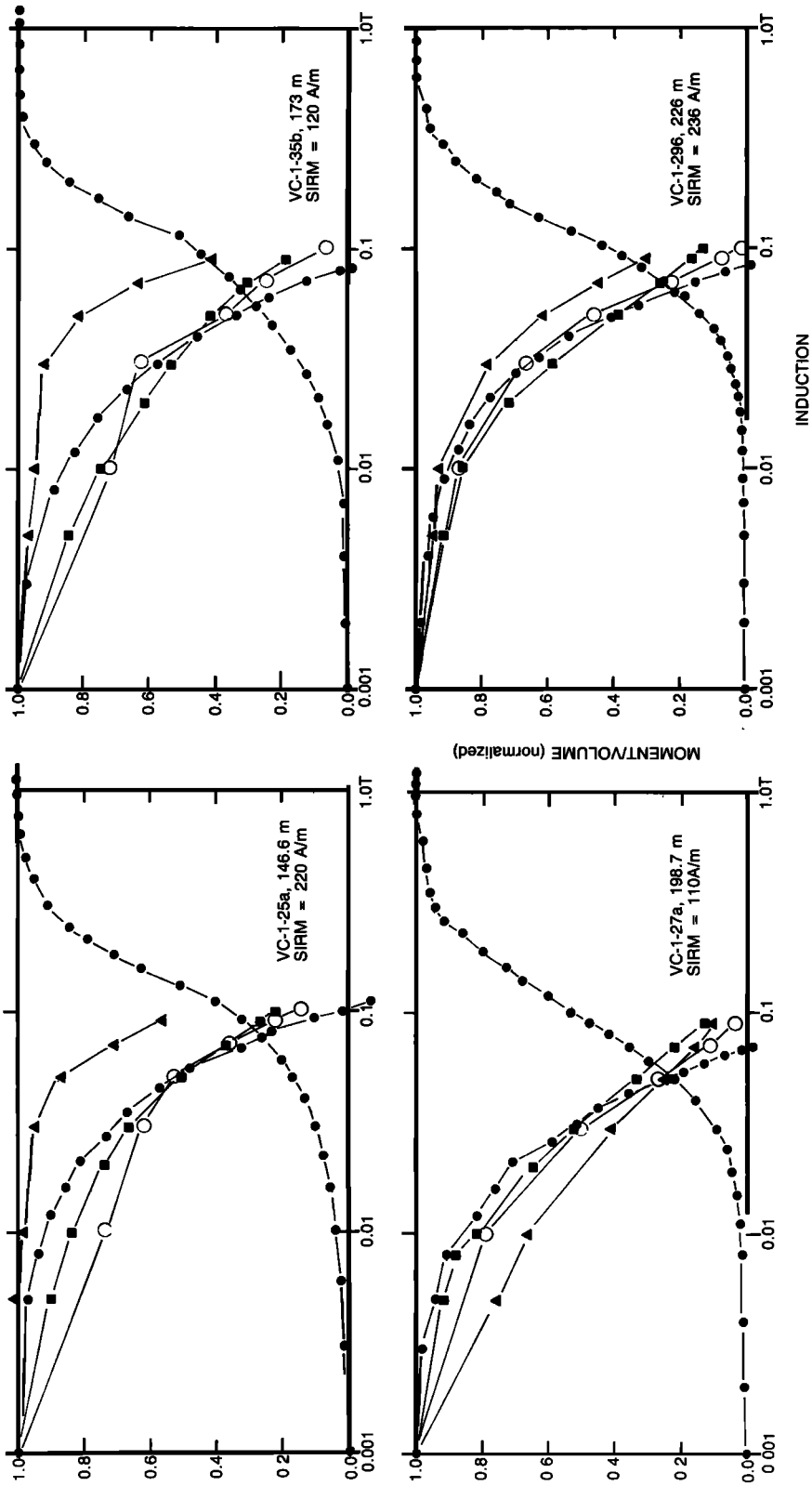



Fig. 7. Semilog plot of normalized magnetization parameters versus magnetization induction for specimens representative of samples of lithologies from the Quaternary volcanic section. Curves denote IRM acquisition and backfield demagnetization of saturation IRM (SRIM) (data points as circles), af demagnetization of SIRM (squares), af demagnetization of NRM (upright triangles), and af demagnetization of ARM (open circles).

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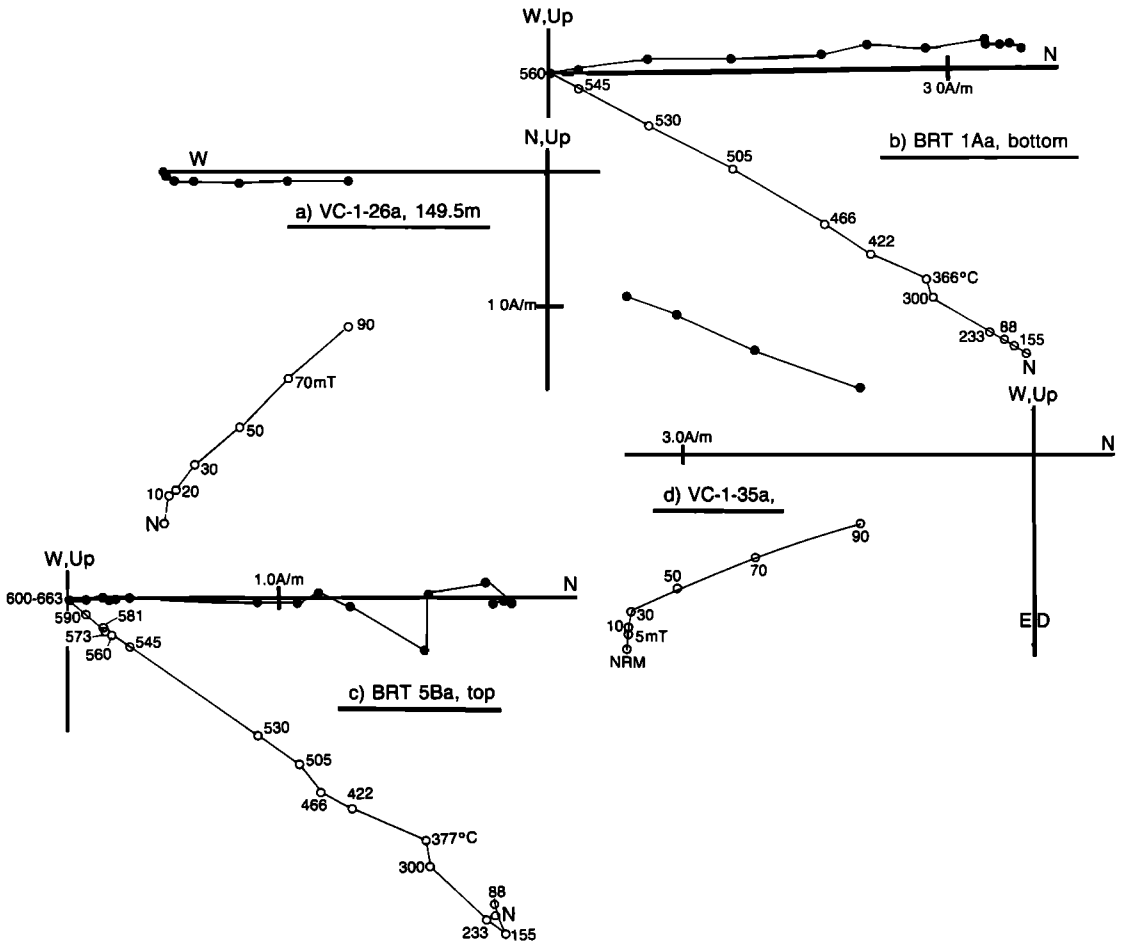



Fig. 8. Orthogonal demagnetization diagrams, Battleship Rock Tuff (from VC-1 as well as surface sites) and VC-1 Rhyolite, arranged in stratigraphic order. Conventions follow those in Figure 6.

Tilting of strata intersected by azimuthally unoriented, vertical core would logically impart a significant dispersion to the inclination of any preexisting RM if the amount of tilting was not recognized. This complexity has been suggested as a possible source of inclination dispersion in layer 2 ocean crust basalts [e.g., *Verosub and Moores*, 1981; *Van Wagoner*, 1983]. Any RM component acquired during the latest Tertiary and Quaternary would clearly postdate tilting, precluding dispersion of inclination data. The paleomagnetic data from the late Paleozoic section have not been corrected for possible tilt of sedimentary strata, although we recognize the slight northeastward dip of these strata in upper San Diego Canyon.

*Abo Formation.* The Early Permian (middle-late Wolfcampian to early Leonardian [*Pray*, 1961; *Otte*, 1959]) Abo Formation was intersected from 335.3 to 432.8 m. Specimens from 27 samples were prepared for measurement. Intensities of NRM ranged from $7.0 \times 10^{-4}$ to $6.0 \times 10^{-2}$ A/m. The Abo Formation consists predominantly of hematitic siltstones and sandstones. The af demagnetization proved ineffective in resolving a well-defined magnetization with an inclination of late Paleozoic affinity. If a significant fraction of the NRM was removed, it was always a moderate positive inclination RM component (e.g., specimens VC1-241 and VC1-80, Figure 10).

Detailed thermal demagnetization afforded the resolution of at least three RM components, often present in the same specimen. The orothogonal demagnetization diagrams in Figure 10 illustrate the complexity of the NRM in this unit. In order of increasing range of $T_{lub}$, the first component removed, if present, was consistently of moderate to steep positive inclination. This component was unblocked by about 300°C and typically constitutes a relatively small percentage of the NRM (e.g., specimens VC1-14 and VC1-B7, Figure 10), although in some cases the component is of greater intensity than higher unblocking temperature components (e.g., specimens VC1-91 and VC1-80, Figure 10). The second component, of moderate negative inclination, was identified in many samples and was unblocked over a range of $T_{lub}$ up to, typically, 550°–600°C. Excellent examples of the progressive unblocking of this component are shown by specimens VC1-11a, VC1-14a, VC1-44, and VC1-87, Figure 10). A third RM component, usually of very shallow negative or positive inclination, is removed up to maximum $T_{lub}$ of pure hematite. In some cases, this component is the only one isolated after the removal of the moderate positive inclination RM at low $T_{lub}$ (e.g., specimens VC1-241 and VC1-243, Figure 10). In others, the moderate negative inclination component is superimposed on the higher $T_{lub}$ RM (e.g., specimens VC1-11a, VC1-14a, VC1-44, and VC1-B7, Figure 10). Inclination data for these components are given in Table 1. As identified on individual demagnetization diagrams, the two high $T_{lub}$ components appear to have similar declinations and partially overlapping $T_{lub}$ spectra. Conse-

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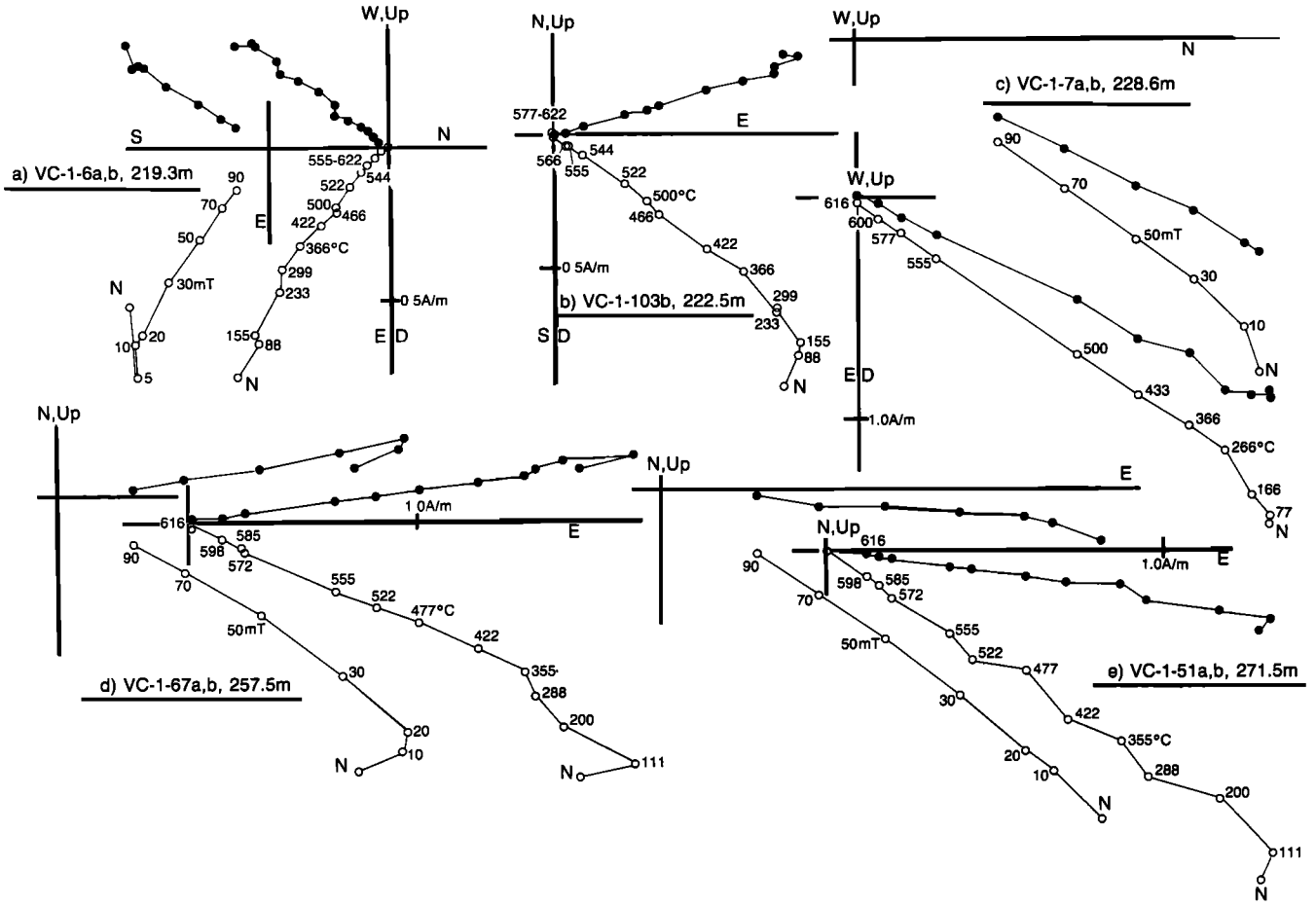



Fig. 9. Orthogonal demagnetization diagrams, VC-1 Tuff, arranged in stratigraphic order. Conventions follow those in Figure 6.

quently, they are often difficult to readily resolve (e.g., specimen VC1-88b, Figure 10). Progressive thermal demagnetization results in a systematic shallowing of the inclination over a temperature range between 500° and 600°C, suggesting the removal of a steeper, negative inclination RM.

Because alternating field demagnetization typically removes all or a considerable portion of the moderate positive inclination RM, this component may be carried by both magnetite and hematite. IRM acquisition curves of Abo lithologies (Figure 11) show that magnetite must be very subordinate in these rocks. The moderate positive inclination RM has a mean inclination of $+52.2°$ ($a_{95} = 7.4°$, $k = 26$), consistent with the possibility that it was acquired during the Brunhes chron, as discussed in more detail below. Assuming a northward declination for this component, then it can be used to reorient core segments [e.g., *Van der Voo and Watts*, 1978; *Kodama*, 1984]. These low $T_{lub}$ components are thus useful for understanding the origin of the higher $T_{lub}$ RMs, which presumably reside in hematite. Where the positive inclination RM could be resolved (i.e., MAD values less than 10°) core segments have been reoriented assuming a north seeking direction. Directions of high $T_{lub}$ RMs, plotted in Figure 12, are consistently south seeking and imply acquisition during reverse polarity chrons. The higher $T_{lub}$ RMs, of shallower inclination, exhibit more southeasterly declinations than those of moderate negative inclination.

*Madera Limestone.* The Madera Limestone was intersected between 432.8 and 802.5 m, and specimens from 105 core segments were prepared for measurement. From paleomagnetic and rock magnetic perspectives this lithologically variable unit is the most complex investigated. Paleontologic studies of the Madera Limestone elsewhere in northern and central New Mexico [e.g., *Myers*, 1973, 1982; *Kelley and Northrop*, 1975] have indicated that the unit ranges from Middle Pennsylvanian (Atokan) to earliest Permian (early Wolfcampian) in age. Strata of the Madera Limestone intersected in VC-1 were deposited in a significantly different environment, probably that of a small basin in the middle or outer shelf (D. Wachs, personal communication, 1986), from sections studied elsewhere in northern New Mexico. The deposition of abundant detrital material shed into the basin as well as formation of hematite, perhaps early in diagenesis, and effects related to late Tertiary igneous activity have all complicated the paleomagnetism of this unit.

Demagnetization behavior characteristic of the Madera Limestone lithologies can be divided into three categories; representative orthogonal demagnetization diagrams are shown in Figure 13. Specimens from over 50% of the core segments are characterized by low NRM intensities ($2.0 \times 10^{-3}$ to $2.0 \times 10^{-4}$ A/m) and NRM directions of moderate to steep positive inclination (Figure 5). Notwithstanding the low NRM intensities, demagnetization is typically straightforward, revealing the univectorial decay of a single RM component (e.g., specimens VC1-165b/165a, VC1-132a/132b, and

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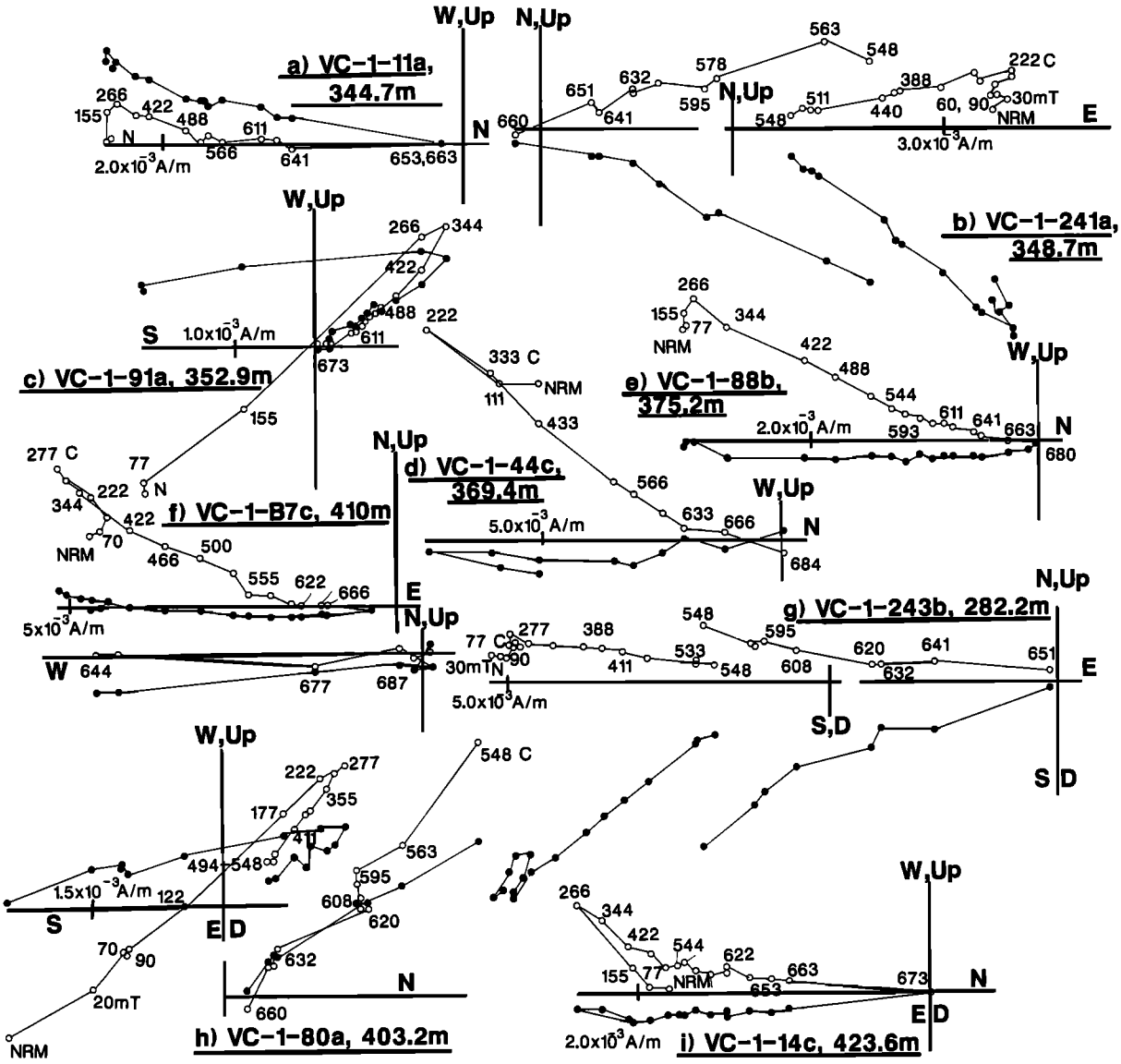



Fig. 10. Orthogonal demagnetization diagrams, Abo Formation, arranged in stratigraphic order. Conventions follow those in Figure 6.

VC1-30a/30b, Figure 13). Median destructive inductions in these cases are typically 10–40 mT. In thermal demagnetization a similar RM component of $T_{lub}$ usually less than 350°C is isolated. This behavior typifies, but is not restricted to, core segments below 588 m.

In the second category, NRM inclinations are more variable and positive and negative. Most importantly, a positive inclination RM is superimposed on a second RM component of moderate negative inclination which often exhibits a declination approximately 180° to that of the positive inclination RM. The negative inclination RM is typically removed in peak inductions up to 100 mT and $T_{lub}$ between 450° and 540°C (e.g., specimens VC1-190c/190a and VC1-158a, Figure 13), suggesting that it, as well as the positive inclination RM, resides in magnetite. Core segments taken from the highly altered portion of the Madera Limestone between 474 and 500 m exemplify this behavior (e.g., specimens VC1-247a/247b and VC1-245a/245b, Figure 13). Alternating field demagnetization has often proved ineffective in fully resolving both components (e.g., specimens VC1-167a/167b and VC1-204a/204b, Figure 13). This behavior implies an overlap of coercivity spectra of the two components yet lack of overlap for the $T_{lub}$ spectra and suggests the positive inclination RM to be of thermoviscous origin.

Finally, the response of many Madera Limestone lithologies to demagnetization indicates hematite as a significant RM carrier. In these cases, NRM intensities are typically 7.0 $\times 10^{-3}$ to 5.0 $\times 10^{-2}$ A/m, and NRM directions are of shallow to moderate negative inclination. Median destructive inductions always exceed 100 mT (e.g., specimens VC1-168a/168b, VC1-160a/160b, and VC1-204a, 204b, Figure 13). That portion of the NRM removed in af demagnetization, or thermal demagnetization at low $T_{lub}$ (up to 350°C), is consistently of moderate positive inclination (e.g., specimens VC1-47a and VC1-204a/204b, Figure 13) and appears to reside in magnetite. Over higher ranges of $T_{lub}$, two RM components are usually isolated. The first, of $T_{lub}$ up to 600°C, is of moderate negative inclination. When clearly defined, the component

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persisting up to maximum $T_{lub}$ for hematite is of shallower inclination and, in a relative sense, more easterly in declination. Specimens VC1-161a, VC1-172b, and VC1-188a (Figure 13), from the interval between 524 and 579 m, offer excellent examples of such behavior which is similar to that exhibited by Abo Formation core segments. Inclination data for these components are given in Table 1.

Again assuming that the moderate positive inclination RM in the Madera limestone lithologies was acquired during the Brunhes (discussed in greater detail below) and is north seeking, the absolute directions of the higher $T_{lub}$ components can be approximated. These directions (Figure 12) are exclusively south seeking; RMs removed over higher $T_{lub}$ ranges are often of more southeasterly declination than those removed at lower temperatures. The RM components of moderate negative inclination have a mean direction of $D = 173.1°$, $I = 46.3°$ ($n = 47$, $a_{95} = 5.5°$, $k = 18.0$; including 16 results from the Abo Formation). It is argued below that the southerly, moderate negative inclination RM was acquired during the Matuyama chron, during development of the Toledo and Valles calderas. The southeasterly, shallow direction of the higher $T_{lub}$ component ($D = 160.1°$, $I = 10.9°$, $a_{95} = 4.4°$, $k = 41$, $n = 28$ samples, including seven from the Abo Formation) yields a paleomagnetic pole at 54.4°N, 106.8°E, reasonably consistent with an age of RM acquisition in the late Paleozoic. Further work on surface material, involving available field tests, is required to assess fully the age of RM acquisition during the late Paleozoic.

*Sandia Formation and underlying rocks.* In north central New Mexico the Sandia Formation is a lithologically variable unit comprising principally terrigenous sediments and minor marine carbonate strata [*Siemers*, 1983; *DuChene*, 1974]. In VC-1 the Sandia Formation is presumably represented by the siliciclastic sedimentary sequence penetrated between 808 and 827 m, stratigraphic details of which are given by *Hulen and Nielson* [this issue]. Beneath 827 m, the VC-1 section consists largely of brecciated and contorted sedimentary rocks. The breccias are of tectonic and/or hydrothermal origin and hematite cements as well as sulfide minerals are common. *Hulen and Nielson* [this issue] proposed that brecciation reflected stress release along the Jemez fault zone during the development of the Valles caldera.

The section of Sandia Formation and underlying rocks is a lithologically complex suite, and the present data, based on specimens from only 14 core segments, may not be entirely representative of the paleomagnetism of these rocks. With the exception of data from core sample VC1-32 (822.5 m) (Figure 14) in hematitic shale, all NRM inclinations and PCA determined inclinations of RM components are moderately positive ($I = +53.9°$, $a_{95} = 4.8°$, $k = 47$, $n = 21$ specimens from 12 samples). NRM intensities range from $2.0 \times 10^{-4}$ to $2.0 \times 10^{-1}$ A/m. For hematite-free strata, af demagnetization is effective in removing a well-defined moderately positive inclination RM component (e.g., VC1-192a, Figure 14). In thermal demagnetization, similar components are removed, with peak $T_{lub}$ of approximately 450°C. The NRM of core segments from the hematitic brecciated zone at about 830 m is clearly dominated by hematite. Moderately positive inclination RM components are effectively removed in thermal demagnetization (e.g., VC1-33b and VC1-22b, Figure 14). In sample VC1-22 the component is removed by 510°C, while in VC1-33 it persists up to at least 670°C. Specimens from core sample VC1-32 (Figure 14) show no indication of a moderately positive inclination overprint. Rather, a shallow positive inclination RM, permissibly of late Paleozoic age, is unblocked above 600°C.

## DISCUSSION

In general, units intersected in the VC-1 experiment all respond well in progressive demagnetization and consequently yield well-defined RMs of geologic significance. Several observations pertinent to this initial phase of paleomagnetic and rock magnetic study on Valles caldera core material are highlighted. Considering the low detail of initial sampling over some intervals where interesting phenomena have been described (typically,. one core sample per 3 m or so), many observations demand future investigation and/or verification and require acquisition of sufficient data from equivalent units exposed on the surface.

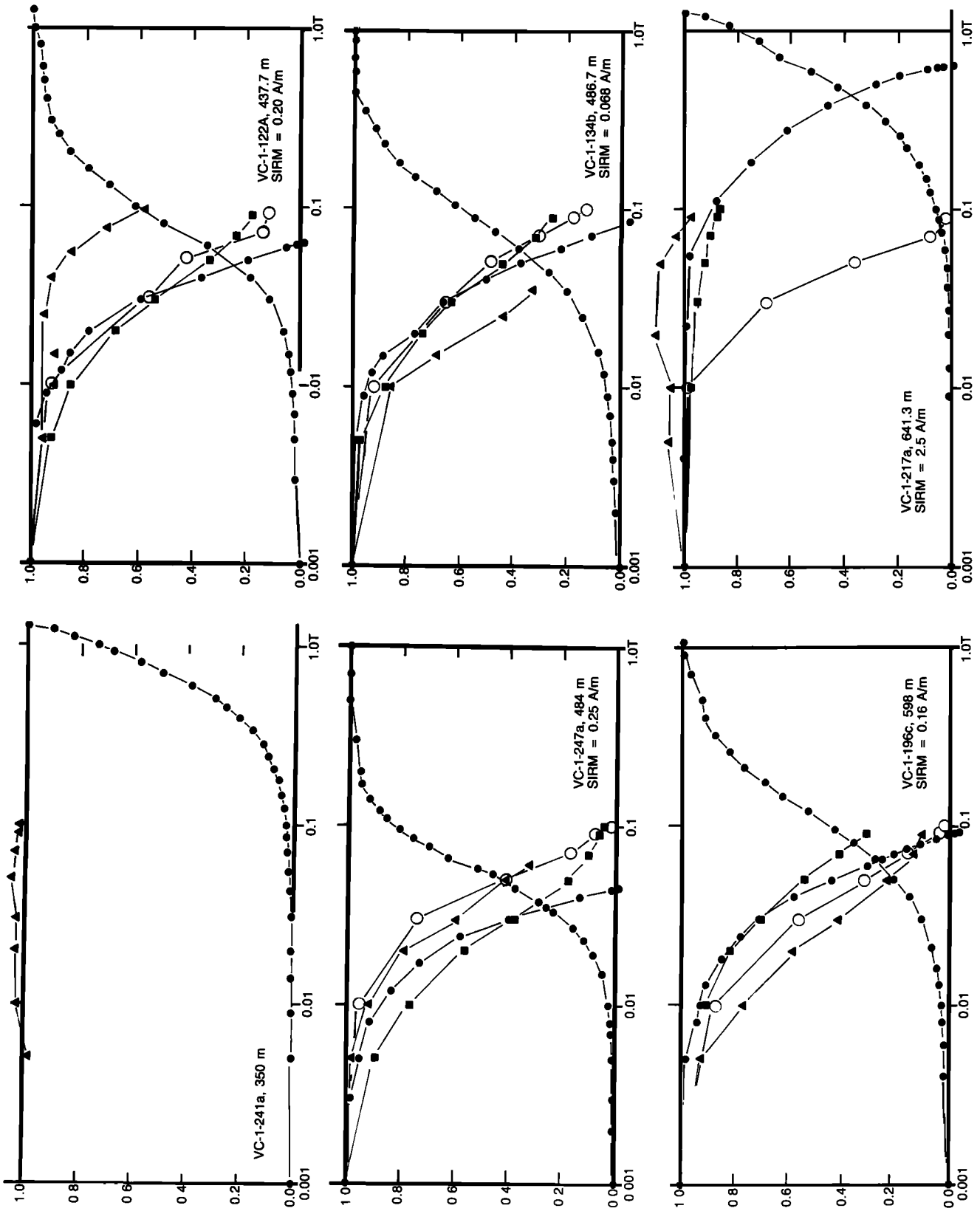
### *Quaternary Volcanic Rocks*

Inclination data from the Quaternary volcanic units studied are consistent with available isotopic age data indicating that the entire VC-1 volcanic sequence was extruded within the last 650 kyr. While demagnetization consistently isolated positive inclination RMs, inferred to be TRMs blocked during normal geomagnetic field polarity, over moderate to high coercivity and $T_{lub}$ ranges, it is noteworthy that negative inclination RM components, of lower coercivity or $T_{lub}$ spectra, are never superimposed on the TRM. Such negative inclination RMs, if acquired at elevated temperatures as viscous overprints (VPTRMs) or chemical overprints (TCRMs), would necessarily imply acquisition, during a reverse polarity chron, after TRM acquisition. RMs isolated from several core segments of poorly welded and lithic-rich Battleship Rock and VC-1 Tuff are of unusually shallow positive to moderately negative inclination. Tentatively, the fact that the RM in these lithologies is not representative of the Brunhes chron is explained as due to a complex NRM (i.e., possibly a composite of randomly oriented RMs of lithics as well as a strongly altered groundmass).

Recognition that the entire volcanic sequence in VC-1 is of Brunhes age has significant implications for study of processes associated with the development of the Valles (~1.12 Ma) and older Toledo (~1.45 Ma) calderas. Of importance to hydrothermal alteration of Tertiary volcanic rocks and, more significantly, underlying strata in the Jemez volcanic field is the fact that the VC-1 volcanic section above the volcaniclastic breccia is unaltered and that the volcaniclastic breccia is only moderately altered [*Goff et al.*, 1986; *Gardner et al.*, 1986]. Substantial phyllic alteration of Paleozoic strata and vein-hosted mineralization, including molybdenite, sphalerite, galena, chalcopyrite, and pyrite, have been noted [*Hulen and Nielson*, 1986, this issue; *Keith*, 1985], prompting investigations into the age of mineralization. Current bottom-hole temperatures in VC-1 (approximately 185°C) are too low for contemporary or near-recent precipitation of such mineral assemblages. These observations led *Gardner et al.* [1986] to propose that hydrothermal alteration of basement rocks in VC-1 may have predated Toledo caldera activity (~1.45 Ma). The significant refinement of the age of the VC-1 sequence allows the possibility, which is strongly supported by paleomagnetic as well as isotopic age data (discussed below), that some hydrothermal alteration in VC-1 resulted from development of the Toledo as well as Valles calderas and may be as young as ~1.0 Ma or less.

Within both the Banco Bonito Obsidian and the VC-1 Tuff,

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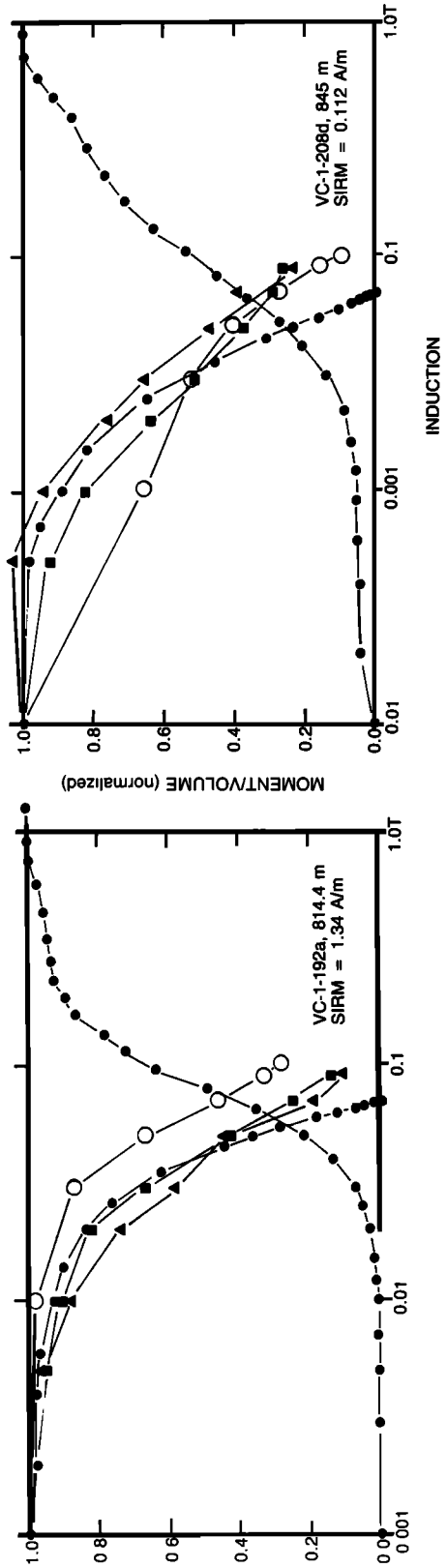



Fig. 11. Semilog plots of normalized magnetization parameters versus magnetization induction for specimens representative of samples from the late Paleozoic section encountered. Curves denote IRM acquisition and backfield demagnetization of saturation IRM (SRIM) (data points as circles), af demagnetization of SIRM (squares), af demagnetization of NRM (upright triangles), and af demagnetization of ARM (open circles).

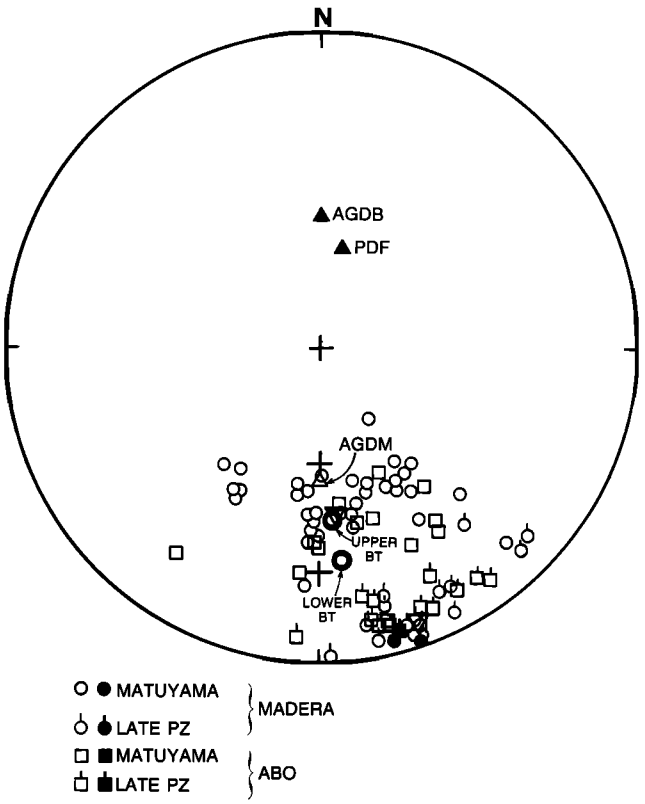



Fig. 12. Equal-area projection of moderate negative and shallow inclination (symbols with tics) RM components of high $T_{lub}$ isolated from the Abo Formation (squares) and Madera Limestone (circles) lithologies. The directions are determined by assuming that lower $T_{lub}$ moderate positive inclination components, when resolvable, are north seeking (i.e., of Brunhes age), allowing azimuthal orientation of the core. Not all negative inclination RMs are shown in Figure 5, as the positive inclination component of low coercivity and/or $T_{lub}$ was not always well defined in demagnetization nor could it always be evaluated using principal component analysis. Axial geocentric dipole field directions for the Brunhes (AGDB) and Matuyama (AGDM) chrons as well as the present-day field direction (PDF) are given by triangles. Unit mean directions (large circles) for the Upper and Lower Bandelier Tuff (BT) are also shown. The large, inverted triangles are the means of the moderate negative and shallow inclination RM components.

subtle yet systematic variations in inclination of RM components, inferred to be TRMs and identified with PCA, are noted. In studying the paleomagnetism of the Tiva Canyon and Topopah Spring members of the Miocene Paintbrush Tuff, southern Nevada, *Rosenbaum* [1986] convincingly argued for considerable dispersion of TRM inclinations in thick (approximately 300 m) tuff sections due to welding processes. RM inclinations of the upper portion of the Topopah Spring member are steep (70°) but flatten to about 20° in the interior portion of the unit and then steepen abruptly at the base. Moreover, on a specimen level in the densely welded, interior portions of the unit, the NRM remaining in thermal demagnetization became progressively shallower in inclination. The two observations imply subblocking temperature deformation in the ash flow sheet, enhanced by high $T_{lub}$ of magmatic magnetic phases, relatively low emplacement temperatures, and substantial stratigraphic thickness. Data from the Banco Bonito (150 m thick) and VC-1 Tuff (120 m thick) intersected in VC-1 may be suggestive of minor amounts of subblocking temperature deformation and TRM inclination flattening occurring in these units. Progressive shallowing of

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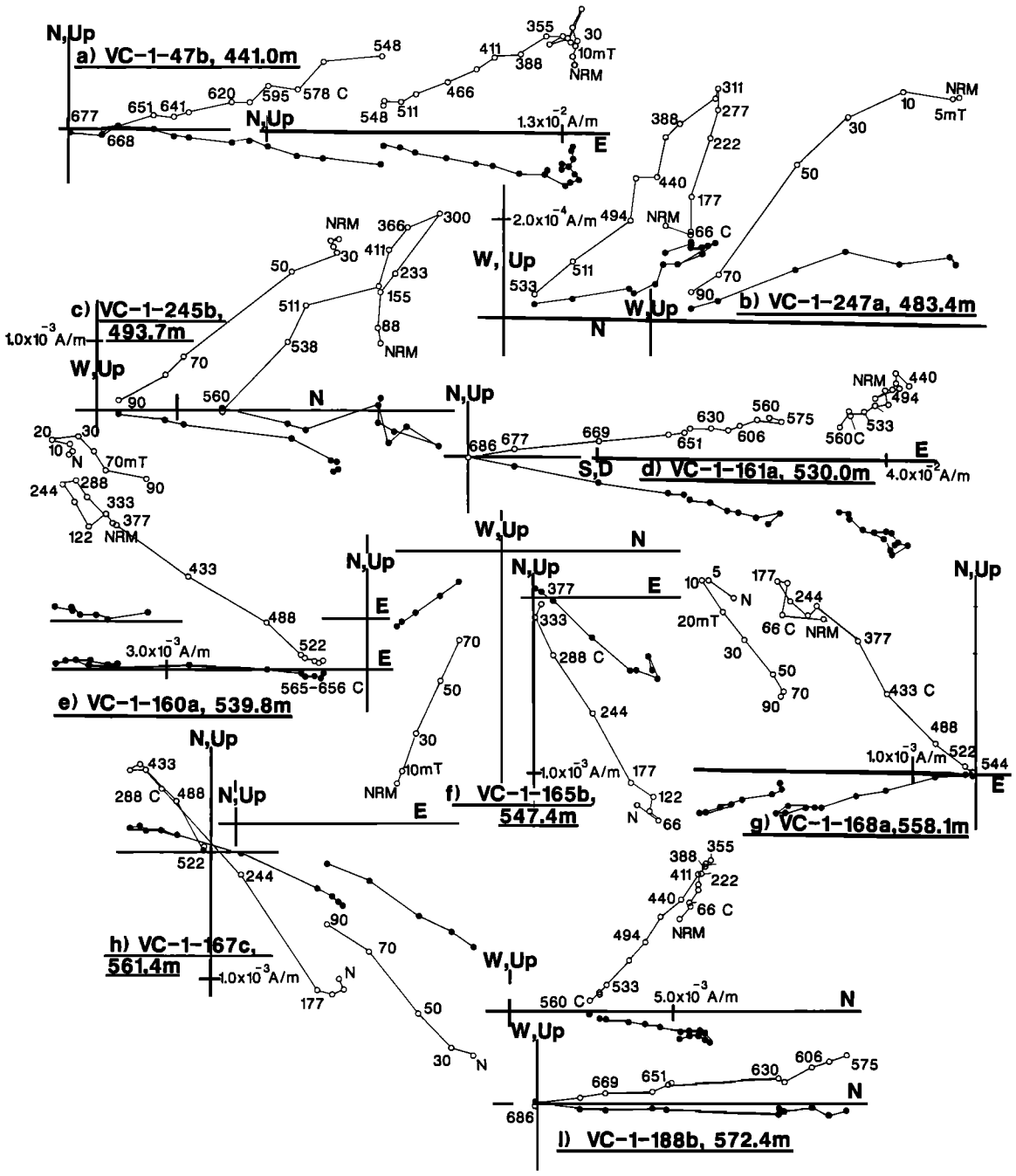



Fig. 13. Orthogonal demagnetization diagrams, Madera Limestone, arranged in stratigraphic order. Conventions follow those in Figure 6.

inclination on a specimen level during thermal demagnetization is not evident (e.g., specimens VC1-104a and VC1-24b, Figure 6, and VC1-103b and VC1-67b, Figure 9). The systematic changes in inclination are not believed to reflect spiraling of the drill stem, based on the one survey determination on the core (discussed above) and the fact that the inclination changes appear related to the geology of the VC-1 volcanic section. In the future, variations in magnetic properties might be correlated with physical properties, but presently only one determination of porosity and density is available for the Quaternary volcanic section [*Dey and Kranz*, 1988].

The limited anisotropy of MS data obtained from the Banco Bonito Obsidian, the VC-1 Rhyolite, and the VC-1 Tuff indicate a well-defined magnetic foliation in these units. Their magnetic lineation may be sufficiently developed to constrain better directions of lava and/or ash flow at the VC-1 site. An anisotropy of MS in volcanic rocks is often readily imparted by preferred alignment of magnetic particles with high susceptibility during flow and compaction [e.g., *Ellwood*, 1982]. Directions of maximum ($k_1$) and minimum ($k_3$) principal susceptibility axes for these units (Figure 15) were calculated by assuming a uniformly directed magnetization in each unit and a due north declination for the TRM isolated during progressive demagnetization. For all three units, inclination of $k_1$ directions are generally shallow, and declinations range from north-northwesterly to northeasterly (AMS data are axial).

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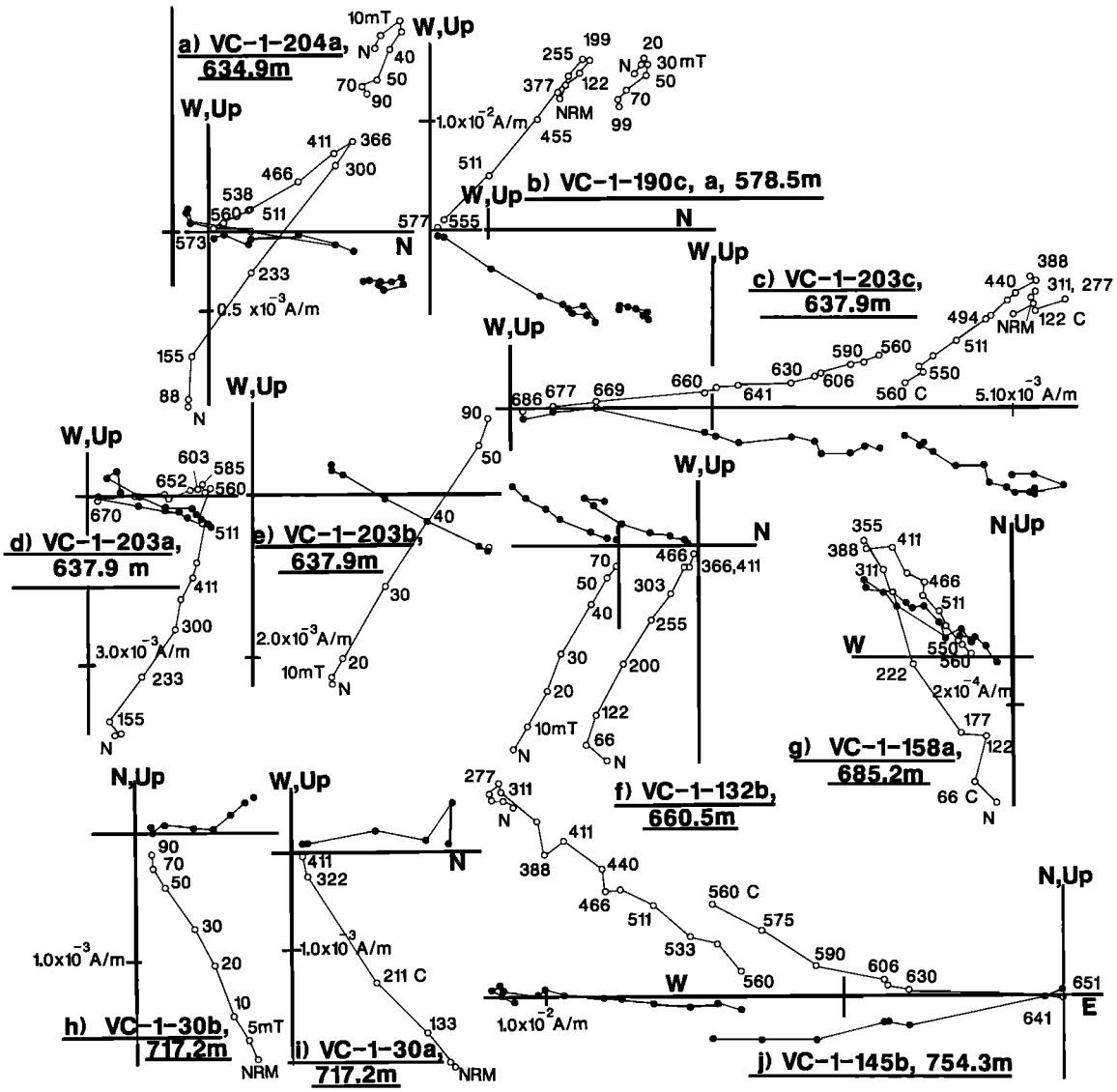



Fig. 13. (continued)

MS foliation planes, perpendicular to $k_3$, are well defined: for the Banco Bonito Obsidian the foliation plane dips moderately eastward; for the VC-1 Rhyolite, steeply westward; and for the VC-1 Tuff, gently eastward. The $k_1$ directions are consistent with field relations (S. Self, personal communication, 1987), suggesting the southward directed emplacement of all three units within a north-northeast to north trending paleo-San Diego Canyon cut into the southwestern moat zone of the Valles caldera.

### *Late Paleozoic Strata and Their Thermally Activated Remagnetization*

The latest Tertiary and Quaternary thermal history of the Valles caldera region has received much attention. The paleomagnetic data from the Abo Formation, Madera Limestone, and Sandia Formation (in particular, magnetizations which appear from their directions to be younger than late Paleozoic) can be explained in terms of the thermal history of the area, and initial research on this subject is briefly summarized. Several studies have examined the response behavior of isotopic dating systems from recovered core of Precambrian rocks of the precaldera "basement" [e.g., *Naeser*, 1981; *Laney and Laughlin*, 1981; *Brookins et al.*, 1977; *Harrison et al.*, 1986]. Estimates of the duration of moderate temperature heating at the Fenton Hill site, based on early work, varied from 1 to about 4 Ma. Very simply, the basement should have experienced one or more periods of anomalously high temperature in response to the thermal development of the caldera and the earlier, extensive Jemez Volcanic Field. To explain the high temperatures observed at the Fenton Hill site, *Kolstad and McGetchin* [1978] invoked the conductive cooling of a large subcaldera magma chamber emplaced at about 1.0 Ma. Their causative heat source extended well beyond the caldera margin and had an upper boundary at a depth of approximately 3.0 km, just below the maximum depth of drilling at the Fenton Hill site but over 2 km below the maximum depth of VC-1, 8 km to the southeast. The $^{40}Ar/^{39}Ar$ age spectra data and revised thermal profiling by *Harrison et al.* [1986] indicated that the most recent Fenton Hill heating event may have been much younger than the main Valles caldera event,

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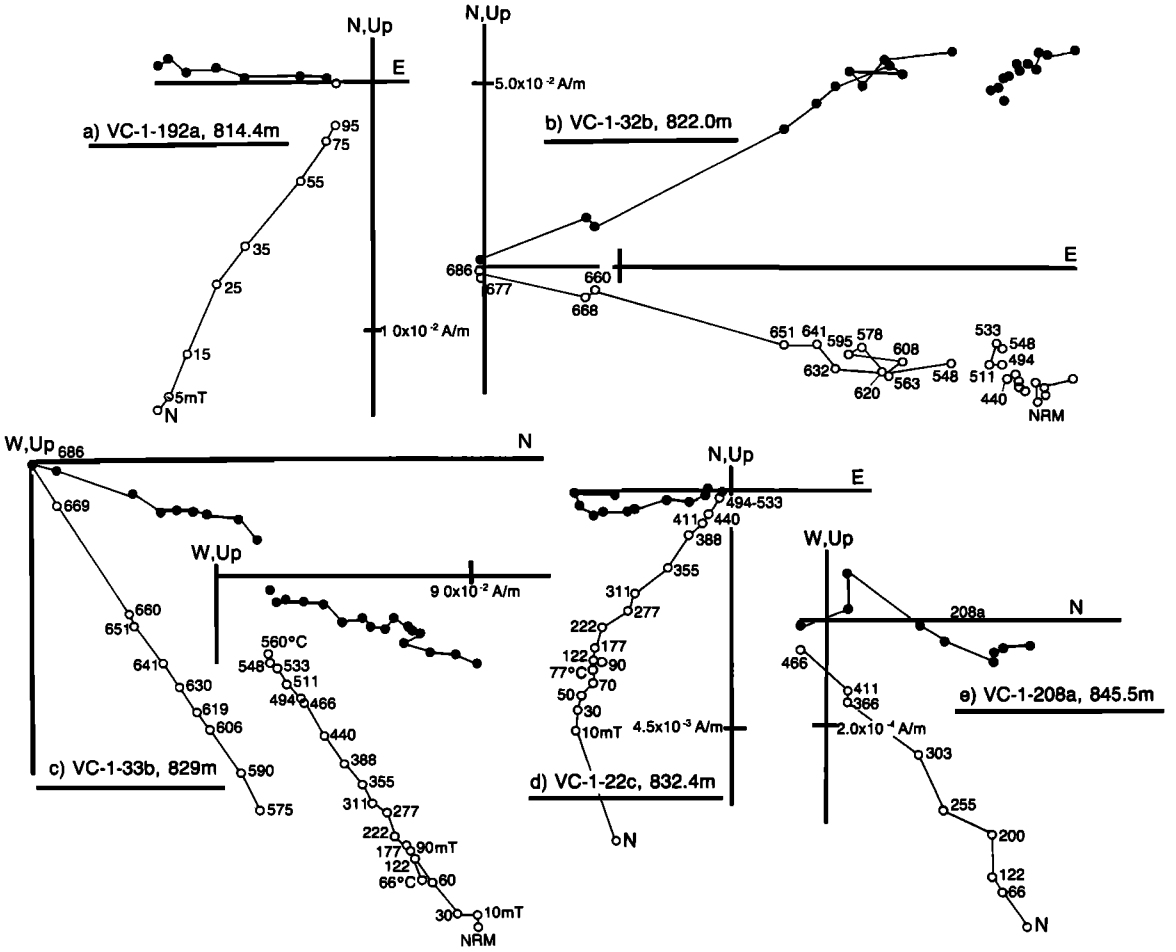



Fig. 14. Orthogonal demagnetization diagrams, Sandia Formation, arranged in stratigraphic order. Conventions follow those in Figure 6.

as had been previously suggested. Maximum estimates of peak heating duration range from 3 to 60 kyr. A shallow thermal disturbance probably related to flow of hot water above the Precambrian unconformity occurred about 10 kyr. Fluid inclusions in quartz and calcite veins in Madera and Sandia strata [*Sasada*, this issue; *Hulen and Nielsen*, this issue] give maximum homogenization temperatures between 270° and 285°C. Veining and attending mineralization/alteration presumably occurred prior to lower-temperature thermal disturbances of late Quaternary to contemporary time at the elevation of VC-1.

Many, typically very detailed, thermal demagnetization experiments were carried out on Abo Formation (Figure 10) and Madera Limestone (Figure 13) core material from VC-1, and

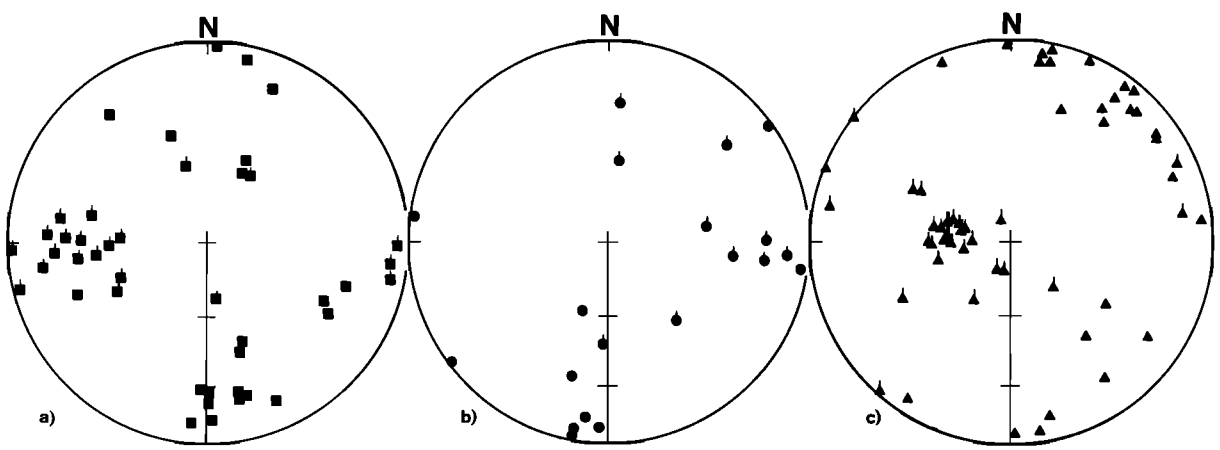



Fig. 15. Equal-area, lower hemisphere projection of maximum ($k_1$) and minimum ($k_3$, symbols with tics) principal susceptibility directions from specimens of (*a*) the Banco Bonito Obsidian, (*b*) VC-1 Rhyolite, and (*c*) VC-1 Tuff. Specimens have been azimuthally oriented assuming a northward declination of the major RM component (TRM) identified in progressive demagnetization.

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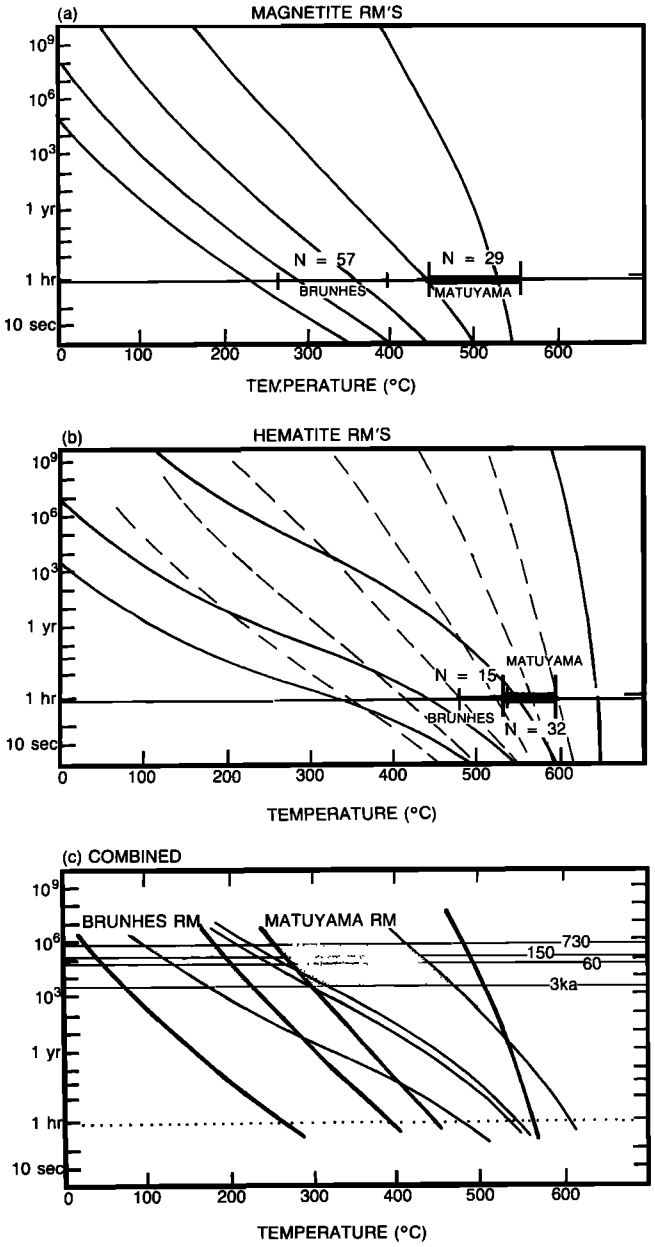



Fig. 16. Thermal activation time versus temperature diagrams showing $T_{lub}$ data for Brunhes and Matuyama age magnetizations in Abo Formation, Madera Limestone, and Sandia Formation lithologies. Solid curves in (*a*) magnetite and (*b*) hematite are from *Middleton and Schmidt* [1982] after *Walton* [1980]. Dashed lines in Figure 16*b* (hematite) are from *Pullaiah et al.* [1975]. The *N* values in Figures 16*a* and 16*b* refer to the number of core segments (samples) yielding laboratory unblocking temperature data. (*c*) The Middleton and Schmidt curves for the two phases (heavy lines, magnetite; light lines, hematite) are superimposed over the range of geologic time appropriate for activation of these Quaternary magnetizations (heavy stipple, Matuyama RM; light stipple, Brunhes RM).

overall, the $T_{lub}$ values of RM components appear consistent with thermal events affecting the VC-1 section in the Quaternary. The two RM components of relatively low $T_{lub}$ of moderate positive and negative inclination (Table 1) are assumed to be geologically young (secondary) and interpreted as thermoviscous RMs, occasionally superimposed on a higher $T_{lub}$ RM of late Paleozoic age. Were these magnetizations of chemical origin (CRMs), then discussion of $T_{lub}$ data in terms of the thermal history of VC-1 is inappropriate, as discussed further below. There is a range in the peak $T_{lub}$ of magnetizations carried in magnetite and hematite. The spread in $T_{lub}$ values may reflect contributions from (1) inaccurate determination of the exact temperatures at which components are removed, as discussed by *Middleton and Schmidt* [1982], (2) obvious lithologic variations and, permissibly, grain size/shape relations in the magnetic carriers in those investigated, and (3) partial contamination of thermoviscous RM components by those of chemical origin. The variations in $T_{lub}$ are probably not strongly influenced by large ranges in the chemical compositions of the RM-carrying phases.

Carbonate rocks of the Madera Limestone, whose magnetization is usually dominated by magnetite, exhibit complete unblocking of a moderate positive inclination RM by 300°–350°C ($n = 57$ core samples, Figure 16*a*). Similarly, an RM residing in part in magnetite in Abo Formation strata is unblocked by 350°C. In the relatively few cases where moderate positive inclination RMs are carried in hematite, however, they are usually removed at $T_{lub}$ ranging from 450° to 550°C ($n = 15$ core samples, Figure 16*b*) but occasionally at $T_{lub}$ greater than 650°C; the wide range of hematite $T_{lub}$ is discussed below. Where identifiable in further thermal demagnetization, a moderately negative inclination RM in both magnetite- and hematite-dominated lithologies is isolated over higher $T_{lub}$ ranges. In magnetite-dominated lithologies the negative inclination RM is unblocked over a temperature range of 450°–550°C ($n = 29$ core samples, Figure 16*a*), while in hematite-dominated lithologies it is generally unblocked between 550° and 600°C ($n = 32$ core samples, Figure 16*b*). For comparison, this moderately negative inclination RM is not observed in pilot thermal demagnetization work on Abo Formation units elsewhere in northern New Mexico. RMs of higher $T_{lub}$ consistently exhibit shallow inclinations, in keeping with a probable late Paleozoic age. Because declinations of the two overprint RMs are approximately 180° apart and that when corrected for the north seeking direction of the positive inclination RM, higher $T_{lub}$, shallower inclination RMs possess declinations of late Paleozoic affinity (see above), the two overprint magnetizations are considered to be geologically significant and are inferred to have been acquired in response to two different thermal events in Quaternary time. While tentative, a working hypothesis is that the positive inclination low $T_{lub}$ RM is of Brunhes age (less than 730 kyr) and was activated during a geologically young thermal event, or one of similar magnitude, identified by *Harrison et al.* [1986] at the Fenton Hill site. A better refinement of the age of the event is impossible; *Goff and Shevenell* [1987] showed that hydrothermal fluids have flowed down the Jemez fault zone for the last 1.0 Ma. The direction of the negative inclination RM is consistent, on the other hand, with thermal activation during the Matuyama chron, during the main Toledo and/or Valles caldera forming events at 1.12 and 1.45 Ma, respectively, and with post-Valles moat rhyolite emplacement to 0.73 Ma [*Gardner et al.*, 1986]. The negative inclinations of the intermediate $T_{lub}$ RMs (Figure 13), in both the Abo Formation (mean $I = -35°$) and the Madera Limestone (mean $I = -47°$) are shallow in comparison with a time-averaged Quaternary axial geocentric dipole field ($I = \pm 55.5°$). Rapid acquisition of the thermal overprint may have occurred. The upper, more voluminous (Tshirege) Member of the Bandelier Tuff was erupted at approximately 1.12 Ma, during a period of anomalously shallow geomagnetic field inclination ($I = -34°$).

In a manner first described by *Middleton and Schmidt*

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[1982], $T_{lub}$ data for each overprint RM carried by magnetite and hematite may allow unique geologic temperature-time solutions for activation of Brunhes and Matuyama age magnetizations in the late Paleozoic strata of VC-1. Evaluation of the consistency of such solutions with respect to independent temperature estimates for Valles/Toledo caldera thermal events during the past 1.45 Ma or so provides an additional test of existing theories describing thermal activation (TA) processes. While thermal activation of magnetizations carried by magnetite and hematite has been a subject of considerable theoretical interest [*Dunlop*, 1973; *Pullaiah et al.*, 1975; *Walton*, 1980; *Enkin and Dunlop*, 1987], experimental application of empirical blocking temperature-relaxation time curves to natural processes has met with limited success [e.g., *Kent*, 1985; *Jackson and Van der Voo*, 1986; *Kent and Miller*, 1987]. *Kent* [1985], *Jackson and Van der Voo* [1986], and *Kent and Miller* [1987] have addressed some of the problems associated with application of the curves, and TA theory is briefly described here.

*Neel* [1955] stated that an assemblage of noninteracting magnetic particles approaches equilibrium with an external magnetic field as exp $(-t/\tau)$, where the relaxation time $\tau$ is given by

$$\tau^{-1} = C \exp (e_0/kT) \tag{1}$$

The energy barrier $e_0$ is that associated with moment reversal in the particle, and the frequency factor $C$ is about $10^{10}$. For magnetite particles with shape anisotropy, $e_0$ is related to grain geometry and proportional to grain volume and the saturation magnetization $J_s$ raised to a particular power (in this case, $n = 2$). *Pullaiah et al.* [1975] derived the relation

$$\frac{\ln (Ct_1)}{\ln (Ct_2)} = \frac{T_2 Js^n(T_1)}{T_1 Js^n(T_2)} \tag{2}$$

for an assemblage of identical single-domain grains. $C$ is the frequency factor, equal to about $10^{10}$; $Js(T)$ is saturation magnetization at that temperature. The relation yields time-temperature conditions ($t_2$, $T_2$) which will activate an RM (viscous partial TRM, or VPTRM) acquired at ($t_1$, $T_1$). Their sets of blocking temperature-relaxation time curves for pure magnetite ($n = 2$) and hematite ($n = 4$) have been applied to several geologic situations [e.g., *Van der Voo and Watts*, 1978; *Van der Voo et al.*, 1978; *Geissman et al.*, 1980; *McCabe et al.*, 1983]. The effects of different distributions of grain sizes and anisotropies of assemblages of magnetic particles was evaluated by *Walton* [1980], who offered a revised relation

$$\frac{\ln (Ct_1)^{3+r}}{\ln (Ct_2)} = \frac{T_2 Js^2(T_1)^{2+r}}{T_1 Js^2(T_2)} \tag{3}$$

which is more strongly time dependent than (2). All terms are as in (2); $r$ is an integer. Differences between the approaches leading to (2) and (3) have been discussed by, among others, *Dunlop* [1986] and *Jackson and Van der Voo* [1986]. Equation (2) assumes the nontemperature dependent part of the activation energy (particle shape and volume) to be constant. Thus magnetization activation of the same particles at different time-temperature conditions is determined. *Walton* [1980] derived (3) by assuming a constant value of the total moment of a population of particles with distributed volumes and anisotropies. The equation relates the time-temperature conditions producing the same total moment but not necessarily in the same grains. Equation (2) appears to be more appropriate for predicting laboratory unblocking temperature conditions for assemblages of both single-domain grains and multidomain grains whose activation is controlled by domain wall displacement.

*Middleton and Schmidt* [1982] constructed TA nomograms for pure hematite and magnetite using (3) for $r = -1$, approximating a log normal grain-size distribution. Investigations of carbonate rocks with magnetite-dominated VPTRMs by *Kent* [1985] and *Jackson and Van der Voo* [1986] indicate that (3) may be a far better empirical predictor of $T_{lub}$ in at least some cases, in particular where magnetite is the carrier of the VPTRM. *Jackson and Van der Voo* [1986] raised the possibility that the success of (3) in predicting $T_{lub}$ for magnetite RMs reflects transdomain VRM [*Moon and Merrill*, 1986] processes in pseudo-single-domain particles. For hematite-dominated VPTRMs, *Kent and Miller* [1987], on the basis of demagnetization data and viscosity experiments, argue that *Walton*'s [1980] approach may not be appropriate and that (2) used by *Pullaiah et al.* [1975] is more relevant.

Laboratory thermal demagnetization conditions which result in the removal of the Brunhes and Matuyama age VPTRMs carried by magnetite and hematite in the VC-1 strata are plotted in Figure 16*c*. In all four cases there is a spread of peak $T_{lub}$, which again may be ascribed to several possibilities. Assuming the Brunhes age RM to be a VPTRM, magnetite $T_{lub}$ of up to 350°C and hematite values of up to 550°C would correspond to acquisition of a VPTRM over a time period well less than 750 kyr at temperatures ranging from 100° to about 150°C (Figure 16*c*), in excellent agreement with VC-1 temperature log data ranging from 100°C at the top of the Abo Formation (330 m) to approximately 185°C within the Sandia Formation (840 m) [*Goff et al.*, 1986; F. Goff, personal communication, 1987]. An accurate age of temperature elevation during the Brunhes cannot be determined. As well, the spread in $T_{lub}$ data for the normal polarity RMs does not allow an estimate of the actual duration of enhanced temperatures; the data are consistent with heating lengths ranging from less than 3 to 730 kyr. It is noteworthy that the intersection of temperature-time curves for the normal polarity RMs also agrees with temperatures extrapolated by the author from predicted thermal profiles from *Harrison et al.* [1986, Figure 8] for the recent thermal perturbation at the Fenton Hill site.

Assuming the Matuyama age RM to be a VPTRM, magnetite $T_{lub}$ ranging up to 550°C and hematite values up to 600°C imply higher temperatures for RM acquisition or impossibly (geologically speaking) long heating times (Figure 16*c*). A thermal event ranging from 280° to 350°C and lasting 150 kyr, the time of reverse geomagnetic field polarity spanning eruption of the Upper Member of the Bandelier Tuff and the onset of the Jaramillo subchron during the Matuyama chron, would result in the observed $T_{lub}$ in magnetite and hematite, according to the *Middleton and Schmidt* [1982] curves. A realistic upper time limit for heating during reversed polarity is ~480 kyr, the time spanning eruption of the Lower Member of the Bandelier Tuff and the onset of the Jaramillo subchron. The range of temperatures predicted is consistent with homogenization temperature data on primary fluid inclusions in hydrothermal quartz veins in the Sandia Formation [*Sasada*, this issue; *Hulen and Nielson*, this issue]. Additional independent data to constrain the thermal history (e.g., conodont color alteration indices for the Madera Limestone) are unfortunately not available. The polarity of this

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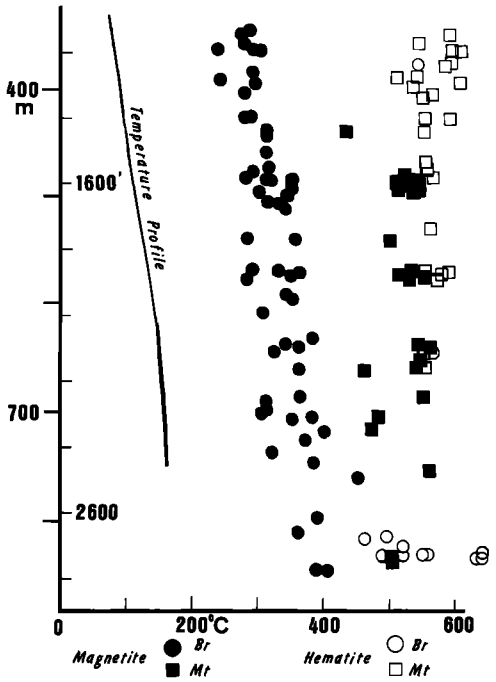



Fig. 17. Plot of $T_{lub}$ data for Quaternary magnetizations versus VC-1 core depth for the late Paleozoic strata. The solid line is the mid-1987 temperature log for VC-1 (F. Goff, personal communication, 1987). Solid circles refer to Brunhes age magnetizations in magnetite, open circles to Brunhes age RM in hematite, solid squares to Matuyama RM in magnetite, and open squares to Matuyama age RM in hematite.

VPTRM is consistent with a 1.0 Ma K/Ar determination obtained on illite/sericite from intensely altered Madera Limestone, at a depth of 500 m [*Ghazi and Wampler*, 1987]. Samples at and near this depth consistently yield a high $T_{lub}$ magnetization, residing in magnetite, of negative inclination (e.g., specimens VC1-247a/247b and VC1-245a/245b, Figure 14).

The $T_{lub}$ data from late Paleozoic strata intersected in VC-1 suggest that much of the section was heated to about 300°C by an initial, high-temperature thermal pulse. Such high temperatures attended alteration and mineralization related to the main Toledo and Valles caldera forming events, beginning at about 1.45 Ma and ceasing prior to 970 ka, at the onset of the Jaramillo subchron. That Matuyama age RMs with $T_{lub}$ less than 500°C or so, regardless of a VPTRM or CRM origin, have been preserved in late Paleozoic rocks requires that these rocks were not affected by a thermal disturbance with temperatures greater than approximately 300°C during the past 730 kyr.

The assumption that all Brunhes and Matuyama age RMs are VPTRMs whose $T_{lub}$ can be applied to TA theory is open to question and demanding of additional rock magnetic and petrographic analysis. Nonetheless, several comments can be made. First, all $T_{lub}$ data used to construct Figures 16*a* and 16*b* are plotted with respect to vertical position of core segments in Figure 17. $T_{lub}$ for Matuyama age RMs in both magnetite and hematite show no uniform variation with depth. The data are consistent with near-uniform heating of the late Paleozoic section, probably through laterally convective fluid circulation out of the Valles caldera down the Jemez fault zone [*Goff et al.*, this issue]. Brunhes age RMs in hematite exhibit variable $T_{lub}$, and some of these, arguably, are in all likelihood of low-temperature TCRM origin (e.g., core segments VC1-33 and VC1-22 in the hematitic breccia zone at approximately 830 m within the Sandia Formation, Figure 14). Remagnetization of hematite-bearing strata is perhaps far more complicated than simple thermal resetting. Recrystallization and coarsening of hematite grains has been shown to occur over geologic time [e.g., *McBride*, 1974; *Folk*, 1976]. The process might necessarily be enhanced by heating and fluid flow.

Brunhes age RMs in magnetite, on the other hand, show an increase in $T_{lub}$, roughly 90°C/0.5 km, through the late Paleozoic section. The $T_{lub}$ observed and their increase with depth are features expected for a lithology having constant magnetic properties heated under the present geothermal gradient in the VC-1 area, according to the *Middleton and Schmidt* [1982] curves. Physical property data for the late Paleozoic section are presently insufficient to permit correlations with magnetic properties. *Dey and Kranz* [1988] report five determinations of porosity and density in the late Paleozoic section. A sample collected at 472 m, in the highly altered portion of the Madera Limestone, has a porosity of 15%, while more typical Madera Limestone from 648 m has a porosity of 0.5%.

Second, use of appropriate temperature-time curves for hematite VPTMs is still under debate and VC-1 data have bearing on these discussions. *Kent and Miller* [1987] argued for and presented laboratory results supporting the *Pullaiah et al.* [1975] curves as more applicable to natural VPTRMs in hematite. Because virtually all geologic hematite is SD [*Stacey*, 1963], the time-temperature evaluations by *Pullaiah et al.* [1975] and *Walton* [1980], which are both based on SD theory, both a priori appear pertinent to hematite. *Enkin and Dunlop* [1987] argue that the low-field approximation theory of *Neel* [1955] is valid for SD hematite but not for SD magnetite, explaining why the *Pullaiah et al.* [1975] time-temperature relationships should fit hematite VPTRMs. With respect to VC-1 data, use of the *Pullaiah et al.* [1975] hematite curves for the Matuyama age RM component, for example, would imply geologic temperatures exceeding approximately 450°C, yet there is no evidence for such a significant thermal disturbance at any point in time. The inference is that either the Matuyama age RM in hematite isolated in both the Abo Formulation and the Madera Limestone is a low-temperature TCRM, which has yet to be proven, or the *Middleton and Schmidt* [1982] curves for hematite are at least occasionally realistic. Perhaps time-temperature relations for geologic hematite are dependent on the grain size distribution of the hematite acquiring a VPTRM, as suggested by *Jackson and Van der Voo* [1986]. These workers reported excellent agreement with $T_{lub}$ for Brunhes age VPTRM in Ordovician carbonates acquired at ambient temperatures by magnetite and hematite using the Middleton and Schmidt curves.

## SUMMARY AND CONCLUSIONS

Paleomagnetic and rock magnetic data from the Quaternary volcanic section intersected in Continental Scientific Drilling Program (CSDP) core hole VC-1 have led to substantial reinterpretations of the volcanic stratigraphy. The fact that the entire volcanic sequence is less than ~650 kyr, as recognized by inclination data from the major units, allows for significant modifications of the geologic history of the portion of the Valles caldera penetrated by VC-1. In particular, alteration and mineralization of the late Paleozoic sequence are no longer temporally constrained to predate development of the

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Toledo and Valles calderas, as petrographically fresh Bandelier Tuff was not intersected in VC-1.

Though azimuthally unoriented, the availability of nearly continuous core sampling as in the VC-1 experiment is extremely attractive from a paleomagnetic and rock magnetic standpoint. When warranted, detailed investigations of variations in rock magnetic properties as well as the nature, direction, and relative significance of RM components are permissible. Directional data for the late Paleozoic section of VC-1 (Figure 12), for example, convincingly illustrate how geologically recent, Brunhes age VPTRMs and, possibly, TCRMs can be used reasonably to accurately orient core, obtaining declination information for geologically older, perhaps more significant RMs. In the case of the VC-1 Paleozoic section, laboratory unblocking temperature information indicates heating of much of the section to temperatures of approximately 250°–300°C (or more) attending development of the Toledo and Valles calderas and their hydrothermal systems. Combined with radiometric age information and fluid inclusion temperatures, the paleomagnetic data can serve as a very powerful tool for detecting and dating thermal perturbations in systems affected by younger igneous/thermochemical activity. Consequently, paleomagnetic and rock magnetic research should be considered integral components of fundamental laboratory investigations in all CSDP coring endeavors.

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